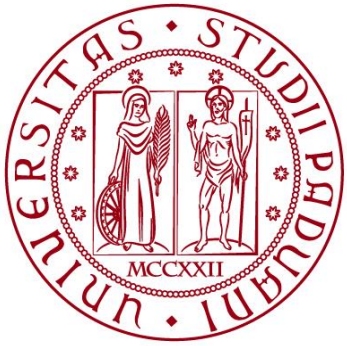
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Corso di Laurea magistrale in Quantitative and Computational Biosciences



TESI DI LAUREA

Modeling SF3B1 Recruitment Kinetics: Branch-Point vs. Decoy Competition under antitumoral treatment

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Abstract

SF3B1 is a core spliceosomal factor that recognizes the branch point during pre-mRNA splicing, a step frequently disrupted in cancer and targeted by spliceostatin compounds. Branch point recognition occurs in competition with nearby non functional sites, and this competition is further shaped by pharmacological inhibitors that block the conformational transition required for stable binding with the SF3B1 complex. However, no existing kinetic framework integrates site competition and drug-induced conformational inhibition into a single, quantitative description. This thesis addresses this gap by developing a four-state kinetic model of SF3B1 recruitment within the linear framework for biochemical systems, with the steady-state occupancy solved analytically. A Compositional Data Analysis (CLR-based) approach is used to normalize the NGS read counts, addressing the compositional nature of the sequencing data. Model parameters are estimated through multistart L-BFGS-B optimization and validated through non-parametric bootstrap. The model achieves a good global fit across both tested drugs (weighted R^2 of 0.84 for Pladienolide B and 0.85 for H3B-8800). However, inspection of the estimated parameters reveals values that are discordant with biological expectations, suggesting that the current framework, while promising, still lacks several steps needed for a fully reliable description of SF3B1 recruitment.

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1 Introduction

1.1 The Splicing Cycle and Complex A

Human genes are interrupted by non-coding regions called introns which are separated by the coding regions, the exons. Before the newly synthesized pre-mRNA can be translated into a protein, the introns must be removed and the exons joined together to form a mature mRNA. This process is called splicing, and it is permitted by the spliceosome, a large and dynamic ribonucleoprotein machine that reassembles on each intron at every ligation. Chemically, splicing occurs through two consecutive reactions, both centered on three short sequences located at the boundaries of the intron: the 5' splice site, the branch point site (BP), and the 3' splice site together with an adjacent polypyrimidine tract (PYT). In the first reaction, a specific adenosine within the branch site attacks the 5' splice site, releasing the first exon and forming a looped intermediate; in the second reaction, that free exon attacks the 3' splice site, joining the two exons and releasing the intron.

The spliceosome does not exist as a pre-formed structure. It assembles step by step on each intron through the ordered recruitment of five small ribonucleoprotein particles, named U1, U2, U4, U5, and U6 snRNPs, together with numerous additional protein factors. This assembly proceeds through a series of well-characterized intermediate complexes, referred to as E, A, B, B^{act}, B*, C, C*, and P (Cretu et al., 2018) where each complex is responsible for a specific structural check or chemical phase of the splicing cycle. Among these steps, the formation of complex A is particularly important for this thesis: it is the stage at which the U2 snRNP, and specifically its SF3b subcomplex, is stably recruited to the branch site, committing the intron to a specific splice site choice.

The same pre-mRNA can be spliced in different ways, including or excluding specific exons, or using alternative splice sites, a phenomenon known as alternative splicing. This process increases the number of distinct proteins that a single gene can produce, and it is strictly regulated in cells. When this regulation is disrupted, for instance by mutations affecting core spliceosomal proteins, splicing can become error-prone generating aberrant transcripts. This is not a marginal issue: dysregulated splicing has been recognized as a contributing factor in a range of human diseases, including cancer, where it can alter the function of proteins that are essential for physiological processes.

Recognition of the branch site is a particularly delicate step of this process, because in humans, unlike in simpler organisms, its sequence is short and only

loosely conserved. Its stable identification therefore depends on a specific protein, SF3B1, and it can be further complicated by the presence of alternative, non-functional branch-site-like sequences nearby, here referred to as decoy sites. These two elements, SF3B1 and the competition it mediates between the branch point and decoy sites, are the central focus of this thesis and are introduced in detail in the following sections.

1.2 SF3B1 in Cancer and Pharmacology

SF3B1 is the most frequently mutated spliceosomal protein in human cancer, with point mutations with the highest incidence in myelodysplastic syndrome (MDS) and chronic lymphocytic leukemia (CLL), and a lower frequency across various solid tumors (Cretu et al., 2016). These mutations do not simply reduce SF3B1 function, they are thought to alter branch site selection itself, favoring the use of alternative 3' splice sites possibly leading to aberrant mRNAs. This suggests that studying SF3B1 can give information from splicing fidelity to cancer, where a single amino acid substitution can be sufficient to redirect the outcome of branch site recognition.

Because of this central role, SF3B1 is also a validated pharmacological target. Several natural compounds such as the pladienolide, spliceostatin, and herboxidiene families, bind SF3B1 directly and interfere with normal branch site usage, altering alternative splicing patterns in a way that is preferentially cytotoxic to malignant cells relative to normal ones (Cretu et al., 2018). Pladienolide B (PB) and synthetic derivatives as H3B-8800 (H3B), are among the compounds investigated for antitumor use, and their structural characterization has clarified how it stalls SF3B1 in a conformation that prevents stable branch site interaction. This combination, a disease driven by altered site selection and a drug class that acts on the same selection mechanism, is what makes SF3B1 a relevant system in which to study branch site competition quantitatively.

However, the structural work in this area remains qualitative: it identifies where and how a modulator binds SF3B1, but it does not explain how the outcome of the competition between the branch point and a decoy site depends jointly on the intrinsic strength of these two elements and on the presence of the drug. What is missing is a framework able to consider, at the same time, the physiological competition between the two cis-regulatory elements and how this competition is reshaped by pharmacological perturbation, since a compound acting on one part of the system could, in principle, shift the balance of site usage even at sites it does not directly bind. This gap, addressed in detail in section 1.7, motivates a model capable of

treating site competition and drug action as two interacting components of the same kinetic system rather than as separate phenomena.

1.3 The Kinetic Competition for the Branch Site

Branch site selection is often presented as an isolated event described simply as a matter of binding affinity between SF3B1 and the intron: the site with the strongest sequence wins. This is a simplification of the cellular environment in which splicing actually takes place. Splicing factors such as SF3B1, are not present in unlimited amounts: they exist as a shared pool that must be distributed across many transcripts at once. At the same time, a single pre-mRNA molecule can contain several BP-like candidate motifs within the same intron, competing for the same limited pool of splicing factors.

These two constraints, a limited pool of factors and multiple candidate binding sites, mean that branch site selection cannot be reduced to a simple question of which sequence binds SF3B1 most tightly in isolation. It is better understood as an active kinetic competition, in which multiple candidate sites compete in real time for the same, shared pool of SF3B1 complexes. In this competition, the site that ultimately prevails is not necessarily the one with the highest raw affinity, but the one that achieves the longest residence time once bound, that is, the one that is most effective at retaining the complex rather than simply capturing it. This retention is shaped by several factors acting together: the intrinsic strength of the sequence itself, additional proofreading-like mechanisms that can reject unstable interactions, and the presence of competing sites nearby.

This distinction may seem subtle, but it is fundamental to how the problem should be approached. It shifts the central question away from "which site binds SF3B1 best?", a question that could in principle be answered by a static structural or biochemical measurement, to a dynamic one: "how does the occupancy of each candidate site evolve as multiple sites compete for the same pool of complexes?". Answering this second question is not something a structural description can do and requires a kinetic model built to capture how competition between sites stabilizes rather than the representation of an isolated interaction.

1.4 The Single-Pocket Constraint

SF3B1 is the largest subunit of the heptameric SF3b complex, a core component of the U2 small nuclear ribonucleoprotein (snRNP) required for branch site recognition during pre-mRNA splicing. Its C-terminal HEAT-repeat

domain folds into a right-handed superhelix, its conformation is maintained by contacts with the other core subunits, SF3B3, SF3B5, and PHF5A. Together with PHF5A and p14, SF3B1 forms a RNA-binding platform: p14 sits at the center of the HEAT superhelix, where it contacts the branchpoint adenosine, while PHF5A contributes to the same binding surface (Cretu et al., 2016).

This architecture has a direct consequence for how competition at the branch site must be modeled. Because SF3B1 provides one composite binding platform rather than several independent ones, it cannot simultaneously accommodate two separate RNA sequences. As a result, if a pre-mRNA contains both the authentic branch point and a nearby decoy site, the two sequences are not free to bind SF3B1 independently: they must compete for occupancy of the same platform. This structural fact is the justification for treating branch point and decoy binding as mutually exclusive states in the model developed in this thesis rather than as two independent binding events.

1.5 The Open-to-Closed Transition

While binding a single sequence, SF3B1 alternates between two distinct conformational states. Crystal structures of the isolated SF3B core show an "open" conformation, in which the terminal HEAT repeats are too distanced to stably bind the U2 snRNA–branch site helix. In contrast, cryo-EM structures of fully assembled spliceosomes reveal a "closed" conformation, in which the terminal HEAT repeats clamps on the U2/BS duplex and accommodate the branchpoint adenosine within a dedicated pocket formed together with PHF5A. The transition between these two states is driven by a hinge located at HEAT repeats H15–H17, around which the H1–H15 module pivots (Cretu et al., 2018). Because the open and closed conformations occupy the same hinge region and cannot coexist, they are mutually exclusive states of the same molecule, not two independent configurations. This is the direct structural basis for representing SF3B1 recruitment with two separate "bound" states in the model, an open reversible conformation one and a closed stabilized one.

1.6 Mechanism of Pharmacological Inhibition

Compounds from the spliceostatin family and their derivatives, act on SF3B1 by binding within the same hinge region, and they do so because they share a common structural core, a conjugated diene, that fits into the same narrow pocket regardless of the specific compound. For PB, this is directly confirmed by its structure: the drug sits inside an hourglass-shaped tunnel made by HEAT

repeats H15–H17 of SF3B1 and two residues of PHF5A, acting as a wedge that keeps the hinge open and physically prevents it from closing.

This matters because the closed conformation is exactly what SF3B1 needs to reach in order to build the pocket that holds the branchpoint adenosine and to stably lock in the U2/BS duplex. By keeping the hinge open, PB blocks this last step indirectly, without competing with the intron for the same physical spot. In fact, the drug and the branch site adenosine end up occupying overlapping regions of the same hinge, but never at the same time, since one belongs to the open state and the other only exists once SF3B1 has closed (Cretu et al., 2018).

It is this separation in state rather than in space, that justifies how drug action is treated in the model developed in this thesis: not as a molecule competing with the branch site for binding, but as something that suppresses one specific transition, the rate at which SF3B1 converts from its open to its closed state.

1.7 The Gap in Current Models

Structural characterizations alone cannot predict how branch site selection responds to sequence variability or pharmacological perturbation, as they capture discrete conformational states without resolving their kinetic rates. While quantitative frameworks exist for early spliceosomal assembly such as the kinetic modeling of U2AF binding at the 3′ splice site (Donovan et al., 2024), which focus the description on factor recruitment to isolated sites.

Existing kinetic frameworks do not incorporate the spatial competition between alternative binding elements and the open-to-closed conformational switch targeted by spliceostatins within a unified formulation. Bridging this gap between site competition and drug-induced conformational inhibition represents the central objective of this thesis.

1.8 Rationale for the Graph-Based Approach

At this point, the system under consideration has taken shape: SF3B1 recruitment can be reduced to a small number of discrete states, an unbound state, a decoy-bound state, and two branch-point-bound states, open and closed, connected to one another by a defined set of transitions. Once the problem is addressed in this way, a possible mathematical representation is using a kinetic graph, where each state becomes a node and each transition between states becomes a directed edge, labeled with the rate at which that transition occurs. This is not just a convenient setup: it is a mathematical object

that captures with some degree of granularity, everything the model assumes about how SF3B1 can move between states.

The reason this representation is useful becomes clear once the quantity that needs to be computed from it is considered. A kinetic graph of this kind can be analyzed within the linear framework (Gunawardena, 2012), a mathematical approach that makes it possible to calculate the steady-state occupancy of every state directly from the structure of the graph itself. In practice it means that, once the graph and its rates are defined, the fraction of time the system spends in each state at equilibrium can be obtained through an algebraic calculation, without having to simulate how the system evolves over time by integrating a system of ordinary differential equations.

This is where the choice of method connects directly back to the nature of the data available for this thesis. The experimental measurements were not collected as a time course tracking how binding builds up minute by minute; they were collected at equilibrium, under a fixed drug concentration as a single snapshot of the system once it has stabilized. Because of this, the quantity that actually needs to be fitted to the data is the steady-state occupancy of each state, not the dynamics of the system to get there. A modeling approach built around solving for transient dynamics would therefore be solving a problem the data cannot actually inform, while introducing unnecessary computational cost in the process. The graph-based linear-framework approach instead is built to answer exactly the question the data can answer: given a set of kinetic rates, what is the resulting steady-state occupancy? This alignment between the mathematical tool and the structure of the experimental data is what makes this approach not just convenient but the appropriate choice for the problem, allowing the underlying kinetic rates to be inferred directly from steady-state measurements, rather than assumed from separate experiments or approximated through numerical simulation.

1.9 Aim of the Thesis

The goal of this thesis is to build and test a kinetic model of how the SF3B1 protein binds to pre-mRNA during splicing. Specifically, the model accounts for two main factors: direct competition between the functional branch point and an upstream decoy site, and drug inhibition that blocks the final locking of the complex. Unlike in simpler organisms, human branch sites vary widely in sequence from one intron to another. A realistic model should capture these differences in sequence strength rather than treating all sites as identical.

This process was described by tracking four key states:

- **Unbound RNA (S_0):** No protein is attached.
- **Decoy-bound (S_D):** The protein is trapped at the non-productive decoy site.
- **Open BP-bound (S_{open}):** The protein binds the branch point in a weak, reversible form.
- **Closed BP-bound (S_{closed}):** The complex locks into a stable, functional conformation.

To represent different sequence strengths, were use two scaling parameters, β and δ, which adjust how fast the protein dissociates from the branch point and the decoy site. Drug effects from two spliceostatin inhibitors (Pladienolide B and H3B-8800) are included by directly slowing down the transition from the open to the closed state.

The project is carried out in three main steps:

1. **Data Normalization:** Applying a Centered Log-Ratio (CLR) transformation to sequencing counts to remove library-size artifacts caused by read redistribution across conditions.
2. **Model Fitting and Parameter Estimation:** Estimating the kinetic parameters using bounded least-squares optimization, combined with multi-start runs and bootstrap resampling to measure confidence intervals and check parameter stability.
3. **Validation Across Drugs and Sequences:** Assessing how well the model predicts the experimental data across different sequence groups, verifying that estimated parameters are biologically meaningful and consistent between the two drugs.

1.10 Hypotheses

The central question this thesis addresses is whether the four-state topology described above, together with two condition-specific parameters modulating branch point and decoy site strength (β and δ), is sufficient to capture the effect of site strength on SF3B1 recruitment and drug sensitivity, or whether this structure is too simple to reflect the underlying biology.

This question is tested by fitting the model independently to two pharmacologically distinct spliceostatin-family inhibitors, Pladienolide B and

H3B-8800, and evaluating whether the estimated parameters are consistent with the expected biological hierarchy of site strength across both datasets.

From this central question, two distinct, testable predictions follow:

- **On the shared kinetic parameters:** because these describe intrinsic properties of the SF3B1 complex rather than sequence-specific effects, their estimates are expected to be consistent between the two independently fitted drug datasets (PladB and H3B), regardless of which compound was used.
- **On the condition-specific scalers:** stronger binding sites (higher-affinity BP or Decoy sequences) are expected to yield lower scaling values, corresponding to slower dissociation and longer residence time, in a way that consistently distinguishes strength classes from one another.

Because these two predictions inspect different parts of the model, they are evaluated separately. If the fitted parameters match these expectations, this supports the idea that the proposed model structure is sufficient.

On the other hand, deviations from the expected behavior such as condition-specific scalers failing to separate strength classes, are not treated simply as a negative result. Instead, they will be critically interpreted. They can serve as diagnostic evidence of intrinsic issues during the modeling phase, such as parameter non-identifiability. Alternatively, they can indicate that the proposed structure is too simplistic to effectively capture the true dynamics of the biological process.

2 Materials and Methods

The experimental data analyzed in this thesis were generated and kindly provided by Suzanne G. Mays and collaborators. All computational pipelines, data normalizations, and mathematical modeling described in the following sections were developed in this work starting from these raw experimental measurements.

2.1 Experimental Design and Data Processing

To study the splicing mechanism under controlled conditions, an *in vitro* spliceosome assembly assay was performed. This approach removes confounding factors such as drug cell-membrane permeability or secondary regulatory effects from accessory splicing factors: while this isolates the biological process and provides a clean signal with reduced technical and biological noise, it also means the experimental observations reflect a simplified, cell-free system.

2.1.1 Library Design and Cis-Element Architecture

A designed mutagenesis library (N=103,239 sequences) was synthesized, combining specific architectural elements to systematically probe the competition between the branch point and the decoy site. Each sequence is approximately 90 nucleotides long and is structured, from the 5' end, as follows (Table 1):

- T7 Promoter: cgTAATACGACTCACTATAGGGc (25 nt), a constant, unmutated sequence required for in vitro transcription by T7 RNA polymerase.
- Decoy Site: a 6 nt upstream competing site lacking the key adenosine (A) required for SF3B1 stable complex formation. Because the complex cannot lock into a functional conformation without this adenosine, spliceosome progression is blocked. Those motifs are designed as 6 discrete mutated variants plus a condition where the decoy site is absent.
- Anchor Region: a structural spacer element with 2 predefined sequence variants derived from literature, used to physically separate the decoy and functional sites.
- Branch Point (BP) Site: a 7 nt motif containing an invariant, functional adenosine (A) at the 6th position. The remaining 6 positions are fully randomized, generating 4^6=4,096 distinct sequence combinations.

- GCG Conjunction: a constant trinucleotide spacer (GCG) linking the branch point site to the polypyrimidine tract.
- Polypyrimidine Tract (PYT): a downstream regulatory element synthesized in two distinct forms, a strong tract (UUUUUCUUUUCUUUU, 15 nt) and a weak tract (UUgaCUUgUUgaCCU, 17 nt).
- Intron-Exon Junction: CAGGGUACGCAU (12 nt), a constant sequence mimicking the downstream 3' splice site boundary.

The strength classification of these cis-elements is not defined identically across the library.

The decoy site strength is set a priori by design: the six variants were generated by systematically mutating the human consensus branch point sequence and removing the reactive adenine at position 6, so that strength decreases proportionally to the number of mismatches relative to the consensus motif.

The polypyrimidine tract strength is defined by design, based on uridine continuity: sequences with long, uninterrupted stretches of uridines display maximum binding affinity for U2AF65, while introducing purines (G, A) disrupts the poly-U tract, reducing binding strength and spliceosome assembly efficiency.

The branch point strength instead cannot be defined a priori, since all 4,096 randomized BP variants are present in the library without a predefined affinity ranking; BP scores are estimated empirically, by validating observed sequence frequencies against theoretical predictions from the computational tool SVM-BPfinder (Corvelo et al., 2010), and the resulting continuous scores are grouped into discrete strength classes, as detailed later in this chapter.

Element	Length	Num. Variants	Strength Definition
T7 promoter	25	1 (invariant)	-
Decoy site	6	7 (6 + no decoy)	design (a priori, based on distance from consensus BP)
Anchor	15	2	-

Element	Length	Num. Variants	Strength Definition
Branch Point	7	4096	empirical + validation (SVM finder)
GCG junction	3	1 (invariant)	-
PYT	15/17	2	design (a priori, based on uridine tract length)
Intron-Exon junction	12	1 (invariant)	-

Table 1 | **Architectural composition and design of the mutagenesis library**. Summary of the sequence elements comprising the ~90 nt constructs designed to probe branch point and decoy site competition. Columns detail the structural element name, length in nucleotides, number of variants, and strength

2.1.2 In Vitro Assembly Assay and Pharmacological Titration

The synthetic mutagenesis library (N=103,239 sequences) was incubated with HeLa nuclear extracts to provide the required spliceosomal machinery. Library aliquots were then separated and treated either with very low amount of durg (concentration of 10^{-11} denoted as control) or with increasing concentrations of two distinct compounds from the spliceostatin family, Pladienolide B (PladB) and H3B-8800 (H3B); the concentration ranges were selected to retrieve comparable ranges of assembly inhibition between the two compounds.

After incubation, assembly reactions were resolved by native gel electrophoresis, and the specific band corresponding to the molecular weight of the SF3B1 complex was excised, isolating only the RNA sequences bound to the target complex.

The bound RNA extracted from this band was sequenced, and read counts were compared against the baseline input library, sequenced immediately following the initial synthesis, to compute the relative enrichment score for each sequence, as detailed in Section 2.1.5.

2.1.3 Experimental Limitations and Modeling Identifiability

This assay measures which RNA sequences end up bound to the SF3B1 complex, but it does not provide positional information regarding the exact

binding site on the sequence, nor the structural conformation of the assembled complex: the experimental readout isolates and quantifies only the sequences that migrate at the molecular weight corresponding to the SF3B1-containing complex, distinguishing only between bound and unbound pools.

To parameterize the mathematical model, it would have been preferable to obtain site-specific resolution, for instance, via crosslinking or footprinting techniques such as CLIP-seq/SHAPE, or conformational resolution to distinguish between open and closed conformation of the SF3B1 complex. In the literature heparin is used to destabilize weak or open pre-mRNA:SF3B1 interactions, which helps distinguish open/weak complexes from closed/stable ones.

This limitation is central to the computational model. The proposed framework aims to describe four distinct molecular states, free RNA, complex bound at the decoy site, complex bound at the branch point in an open conformation, and complex bound at the branch point in a closed, competent conformation. Because the experimental readout distinguishes only between bound and unbound pools, the open-to-closed transition cannot be directly measured or fitted from these data alone. This is an intrinsic limit on model parameter identifiability, which is discussed further when interpreting the fitted condition-specific scalers.

2.1.4 Sequencing Data Structure and Quality Assessment

The dataset consists of sequencing read counts for each library sequence, measured both in the input pool (pre-selection, immediately following synthesis) and across all output conditions (untreated control and each drug concentration point), which serve as the basis to compute the enrichment scores.

The resulting count matrix contains 103,239 rows, each representing a unique RNA sequence variant with its specific architectural combination (decoy variant, anchor type, branch point sequence, and polypyrimidine tract strength), while the columns correspond to the sequencing libraries: the input reference sample, the untreated control condition, and the titration series for both drug treatments. Each entry in the matrix contains the raw sequencing read count reflecting the abundance of that specific sequence variant recovered from the target gel band under a given experimental condition.

Although the original experimental protocol describes n=3 biological replicates, the sequencing matrix provided contains only a single value (n=1) per condition; without knowing whether those counts represent an aggregated

average or a single sequenced run, the data was treated as a single replicate series.

Despite this limitation, the dataset retains high statistical power at the architectural level: each background (combining decoy variant and PYT strength) is tested across all 4,096 branch point variants, allowing more reliable averaging within each sequence group. Furthermore, in the baseline input library, more than 90% of designed sequences were recovered with at least 10 reads, confirming adequate complexity and sequencing depth.

2.1.5 Binding Enrichment Quantification

To quantify the assembly efficiency of each sequence, the experimental dataset provides an enrichment score based on the DimSum pipeline (Faure et al., 2020). For a given sequence s, its read count in the post-selection output is normalized by the total output library depth, and then divided by its corresponding read count in the pre-selection input pool normalized by the total input library depth; taking the natural logarithm of this ratio yields the log-enrichment score:

$$\mathrm{ES}_s = \ln\left(\frac{c_{s,\mathrm{bound}}/N_{\mathrm{bound}}}{c_{s,\mathrm{input}}/N_{\mathrm{input}}}\right)$$

where $c_{s,\mathrm{bound}}$ and $c_{s,\mathrm{input}}$ denote the raw read counts of sequence s in the bound fraction and in the initial input library, respectively, and N_{bound} and N_{input} denote the total sequencing depth of the bound and input libraries. This score measures whether a sequence is positively enriched or depleted in the SF3B1-bound complex relative to its synthesis abundance.

2.2 Exploratory Data Analysis and the Compositional Constraint

Before applying any normalization to the raw read counts, an Exploratory Data Analysis (EDA) was performed to evaluate whether the raw sequencing data is affected by technical artifacts related to Next-Generation Sequencing (NGS) constraints.

In Figure 1 and 2, the x-axis represents the increasing dose of the spliceosome modulators (from baseline Control to the highest drug dosage), while the y-axis shows the proportional contribution (%) of raw reads. Figure 1 display the stratification of reads across Polypyrimidine Tract (PYT) strengths and its

changes in the whole library due to the drug effect. Figure 2 similarly displays these shifts across the distinct Branch Point (BP) strength (Strong, Intermediate, Weak).

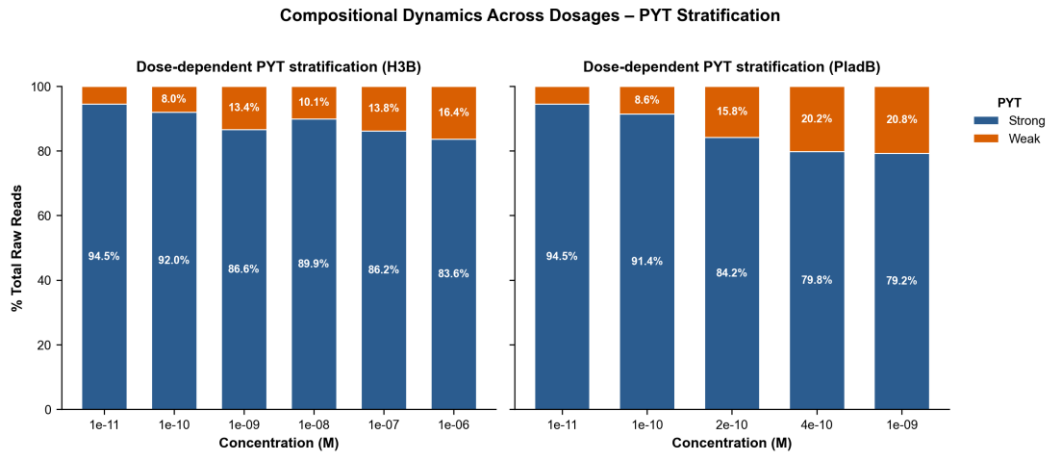


***Figure 1* | Compositional dynamics of Polypyrimidine Tract (PYT) variant populations under spliceosome modulation.** Proportional representation of raw reads for PYT Strong and PYT Weak classes across increasing drug doses. The trajectories illustrate the heavy baseline dominance of PYT Strong reads, which dilutes while the PYT Weak class relatively expands.

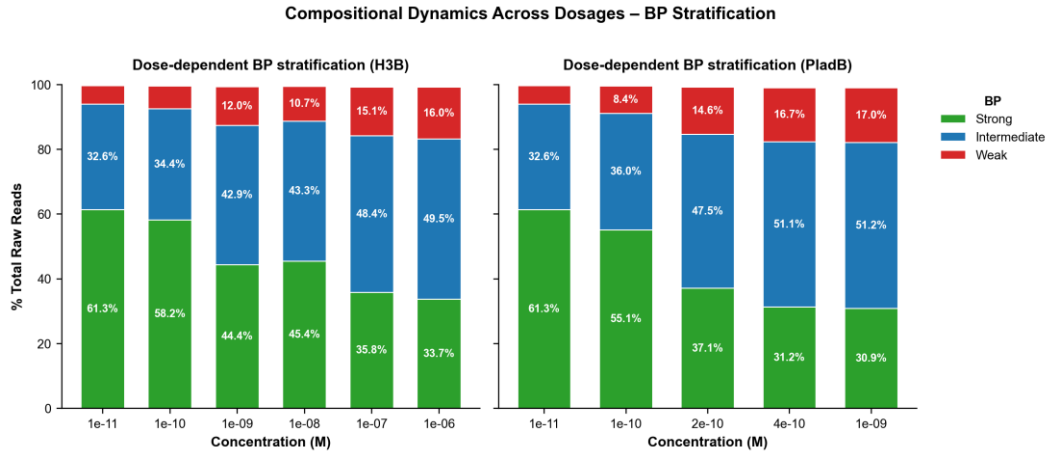


***Figure 2* | Compositional dynamics of Branch Point (BP) variant populations under spliceosome modulation.** Proportion of raw reads mapped to Strong, Intermediate, and Weak categories across increasing concentrations of the modulators (from baseline Control to maximum dose). The plots visualize how the baseline polarization shifts as raw counts redistribute.

As illustrated in both plots, the baseline library (Ctrl) is highly polarized: the *BP_Strong condition* accounts for 61.3% of total raw reads, while *BP_Intermediate* and *BP_Weak* occupy significantly smaller fractions; similarly, *PYT Strong* accounts for 94.5% of total reads. This is coherent with the experimental design, since the sequenced reads correspond to sequences

bound to the complex, so cis-elements favouring binding (*PYT Strong, BP Strong*) are more enriched in the control condition.

Upon application of the spliceosome modulators (H3B and PladB), a strong reduction of *BP_Strong* reads is observed: under Pladienolide B, the *BP_Strong* fraction drops from 61.3% to 30.9% at the highest dosage (PB8), coupled with a +290% relative expansion of the *BP_Weak* condition (Table 2). This inverse trend is confirmed by a strong negative correlation between the Δ Reads of the two classes (Pearson $r = -0.7924$, $p = 0.110$; Spearman $\rho = -0.7000$, $p = 0.188$); the non-significant p-values likely reflect the limited number of drug points ($n = 4$), but the magnitude of both coefficients still suggest an inverse relationship between the dominant and weakest sequence populations.

PladB: Delta vs Baseline				
	Δ Reads (Strong)	**Δ% (Strong)**	**Δ Reads (Weak)**	**Δ% (Weak)**
Dose				
Ctrl	+0	+0.00%	+0	+0.00%
PB1	-562,256	-10.47%	+139,057	+44.32%
PB2	-799,293	-14.89%	+543,468	+173.20%
PB4	-892,250	-16.62%	+821,536	+261.82%
PB8	-685,217	-12.76%	+913,991	+291.29%

***Table 2* | Raw read variations and percentage shifts stratified by Polypyrimidine Tract (PYT) strength.** Absolute differences (*Δ* Reads) and relative percentage changes of raw read counts compared to the preceding condition, stratified by PYT strength classes (Strong versus Weak) across drug titration.

This inverse dynamic may reflect a limitation of sequencing data: because total library depth is fixed, the collapse of a highly abundant class (*BP_Strong, PYT Strong*) mechanically redistributes reads onto the remaining sequences, making it impossible to distinguish true biological signal from this compositional artifact (Quinn et al., 2019). The issue is especially critical here because *PYT Strong* and *PYT Weak* sequences differ in baseline binding by up to two orders of magnitude, so any apparent increase in reads for one group could equally reflect biological recruitment or technical redistribution.

Standard correction methods do not work for this problem: the experimental setup lacks spike-in controls, which would provide an absolute scaling reference; simple approaches like total count normalization or CPM are distorted by the dominant, highly-bound sequences; and standard bulk RNA-seq methods (DESeq2's median of ratios, edgeR's TMM) assume that the majority of features (70–80%) remain invariant across conditions, an assumption violated here by the global shifts induced by pharmacological inhibition. It was then adopted a Compositional Data Analysis (CoDA) framework based on the Centered Log-Ratio (CLR) transformation (Quinn et

al., 2019), which normalizes each count against the sample's geometric mean, preserving relative ratios between sequences, removing dependence on library size, and reducing the read-redistribution artifact.

2.3 CoDa Analysis

The Centered Log-Ratio (CLR) approach expresses each sequence value as the log-ratio of its count relative to the geometric mean of the entire sample. The geometric mean is used because sequencing counts operate on a multiplicative, rather than additive, scale, so it reflects central tendency without being dominated by a few highly abundant sequences

For a sample $\mathbf{x} = (x_1, x_2, \dots, x_D)$ containing D unique sequences, the geometric mean $g(\mathbf{x})$ is defined as:

$$g(\mathbf{x}) = \left(\prod_{j=1}^{D} x_j \right)^{1/D} = \exp\left(\frac{1}{D} \sum_{j=1}^{D} \ln x_j \right)$$

The CLR coordinate for the i-th sequence is then computed as:

$$\mathrm{CLR}(x_i) = \ln\left(\frac{x_i}{g(\mathbf{x})} \right) = \ln(x_i) - \frac{1}{D} \sum_{j=1}^{D} \ln x_j$$

The geometric mean is computed independently within each condition (control or a given drug titration point), and each sequence's log-ratio is evaluated relative to that condition-specific mean.

This guarantees scale invariance ($\mathrm{CLR}(c \cdot x_i) = \mathrm{CLR}(x_i)$ for any constant c), so differences between CLR values reflect true relative shifts in the sequence pool rather than fluctuations in sequencing depth.

The next step is identifying an invariant set of reference sequences, those whose relative abundance remains stable across the drug perturbation. Standard tools for this (ALDEx2, ANCOM-BC) require biological or technical replicates to model variance, which the single-replicate design (n=1 per drug point) does not allow; an alternative criterion was then proposed.

2.3.1 Stability measure and reference set selection

The across-condition variance of the CLR values was used as a proxy for invariance: a sequence with low CLR variance across the drug series has a stable ratio to the geometric mean, making it a good invariant-set candidate.

For sequence i across K conditions:

$$\sigma_i^2 = \frac{1}{K-1}\sum_{k=1}^{K}\left(\mathrm{CLR}_{i,k} - \overline{\mathrm{CLR}_i}\right)^2$$

This criterion carries an element of circularity, since low variance could reflect either true biological invariance or apparent stability induced by compositionality in the library. Anyway given the single-replicate constraint, this empirical variance-minimization approach is the most consistent option available.

To reduce the impact of this bias, two validation strategies were applied: Leave-One-Condition-Out (LOCO) cross-validation by reidentifying the invariant set after iteratively excluding one drug condition, to verify results are not driven by a single outlier concentration; and cross-compound intersection, retaining only sequences identified as invariant under both PladB and H3B, to obtain a reference set independent of the specific drug used.

To select the exact number of invariant sequences without an arbitrary threshold, the Kneedle algorithm was applied (Satopaa et al., 2011), which detects the "elbow" of the variance curve in a data-driven way. The CLR variances are sorted in ascending order and log-transformed (to prevent high-variance outliers from dominating), then both indices and log-variances are normalized to [0,1]. The elbow point i^* is identified as the index maximizing the perpendicular distance $d_i = x_i^{\mathrm{norm}} - y_i^{\mathrm{norm}}$ from the diagonal reference chord connecting (0,0) to (1,1):

$$i^* = \arg\max_i(d_i)$$

Finally, the invariant reference set S_{inv} is defined as all sequences up to the identified elbow cutoff $S_{\mathrm{inv}} = \{s_{(1)}, s_{(2)}, \ldots, s_{(i^*)}\}$. Sequences in S_{inv} exhibit the lowest variance across the drug series and constitute the reference set for normalization.

Applied independently to both treatments, this resulted in $S_{\mathrm{inv}}^{\mathrm{PladB}}$ (7,504 sequences) and $S_{\mathrm{inv}}^{\mathrm{H3B}}$ (7,177 sequences), each ~7.5% of the library.

Their intersection, $S_{\mathrm{core}} = S_{\mathrm{inv}}^{\mathrm{PladB}} \cap S_{\mathrm{inv}}^{\mathrm{H3B}}$ retains ~25% of each individual screen (1.8% of the total library), a small but sufficient set, since the sequences

invariant under both drugs provide a more robust, drug-independent baseline for computing reliable scaling factors.

2.3.2 Normalization

For each condition j, the scaling factor is the geometric mean of the invariant sequences counts within that condition (geometric, again, to prevent extreme counts from dominating):

$$\mathrm{SF}_j = \left(\prod_{i \in S_{\mathrm{core}}} x_{i,j} \right)^{1/|S_{\mathrm{core}}|}$$

Raw counts are then normalized as $\tilde{x}_{i,j} = \frac{x_{i,j}}{\mathrm{SF}_j}$ and the $\log_2$ Fold Change (LFC) between each drug condition j and the untreated control ($j = 0$) is computed as:

$$\mathrm{LFC}_{i,j} = \log_2 \left(\frac{\tilde{x}_{i,j}}{\tilde{x}_{i,0}} \right)$$

With the untreated control serving as the baseline representing physiological binding in the absence of drug, the $\log_2$ FC directly quantifies the drug-induced increase or decrease in assembly efficiency for each sequence.

2.4 Branch Point Strength Classification

To model the competition between binding sites, an objective quantification of relative site strength is required. While the decoy site strength is set *a priori* based on the distance from the human consensus branch point, quantifying the functional strength of the branch point (BP) is less direct because its six flanking positions (of the functional Adenosine) were fully randomized.

Since raw sequencing counts reflect the combined influence of multiple features (BP sequence, PYT strength, decoy competition, and drug concentration), a dedicated three-step strategy was implemented to isolate the binding strength of each branch point:

1. **Isolation of the Baseline Control:** Extraction of a clean reference background to measure empirical read fractions.
2. **In Silico Validation:** Comparison between empirical fractions and theoretical scores from SVM-BPfinder.

3. **Empirical Quartile Classification:** Classification of the validated distribution into discrete categories for modeling.

2.4.1 Isolation of Baseline Control

Read counts in high-throughput assays are driven by multiple co-occurring elements and cannot be used directly as a proxy for BP strength. To eliminate these confounding effects, the reference condition was defined as:

- **No Decoy:** to remove upstream competitive binding.
- **Strong PYT:** to ensure assembly efficiency is driven primarily by branch point recognition rather than weak U2AF recruitment.
- **Control (No Drug):** to exclude pharmacological disruption of the complex.

Under this isolated reference condition, the empirical strength of each individual BP variant (i) is defined as its normalized read fraction relative to the total reads in the reference pool:

$$f_i = \frac{\tilde{x}_{i,\mathrm{ref}}}{\sum_{j=1}^{N_{\mathrm{BP}}} \tilde{x}_{j,\mathrm{ref}}}$$

This fraction captures which branch point sequences are most efficiently assembled when all other architectural and chemical perturbations are removed.

2.4.2 In Silico Validation

To verify that the empirical baseline fraction f_i reflects real affinity, this baseline fraction has been benchmarked against SVM-BPfinder (Corvelo et al., 2010). SVM-BPfinder uses a support vector machine trained on natural human introns, combining position-weight matrices of the branch site with flanking features such as polypyrimidine tract length, tract score, and distance to the 3' splice site.

The sequences were prepared by removing the T7 promoter and trimming immediately after the canonical 3' splice site junction (CAG). SVM-BPfinder was executed in Python 3.12 with the following parameters:

- Species flag: -s Hsap (human model).
- Search window: -l 100 (scanning up to 100 nt upstream of the 3' splice site because the library sequences are approx 90nt).

- Minimum distance: -d 10 (enforcing a minimum 10 nt spacer to account for the AG Exclusion Zone).

Because SVM-BPfinder outputs predicted 9-mers with an associated SVM score, the predictions were filtered to retain only those containing the exact empirical 7-mer motifs. This filtering matched 1,414 out of the 4,096 randomized variants.

This reduction is expected: the synthetic library samples the complete space of all possible 6-mer combinations, including biologically implausible sequences, whereas SVM-BPfinder is trained on evolutionarily conserved functional introns and therefore assigns low scores to motifs that deviate from natural branch site architecture.

The correlation between predicted SVM scores and empirical read fractions was evaluated across the drug points:

- **Pearson correlation (r):** Ctrl = 0.38, PB1 = 0.40, PB2 = 0.45, PB4 = 0.43, PB8 = 0.41
- **Spearman rank correlation (ρ):** Ctrl = 0.55, PB1 = 0.53, PB2 = 0.55, PB4 = 0.48, PB8 = 0.46

The moderate positive correlation suggest that the baseline empirical fraction is considering branch site strength.

2.4.3 Empirical Quartile Classification

To implement the kinetic competition model, the continuous distribution of empirical fractions was discretized into three operational categories using quartile thresholds:

- **Strong BP:** Top 25% ($f_i \geq Q_3$)
- **Intermediate BP:** Central 50% ($Q_1 \leq f_i < Q_3$)
- **Weak BP:** Bottom 25% ($f_i < Q_1$)

This 25%–50%–25% partition separates extreme architectural behaviors (high-affinity vs. low-affinity binders) while maintaining a balanced intermediate baseline. The resulting labels were mapped across all library sequences into the *BP_class* feature.

2.4.4 Biological Coherence Quality Check

To confirm the validity of this empirical grouping, two consistency checks were performed:

- **Consensus Sequence Positioning:** The ranking of the canonical human consensus branch point motif (TACTAAC) was verified. The consensus sequence ranked in the top 6th percentile of the baseline distribution, confirming that high empirical fractions capture optimal binding.
- **Monotonicity Across Conditions:** The mean read abundance of each class across the drug titration series was tracked to confirm that the hierarchy was preserved:

$$\overline{\text{Reads}}_{\text{Strong}} > \overline{\text{Reads}}_{\text{Intermediate}} > \overline{\text{Reads}}_{\text{Weak}}$$

This ranking remained consistent across all drug concentrations, validating the stability of the classification.

2.5 Mathematical Modeling Framework

With the strength classes validated, the next step is to translate this classification into a quantitative modeling framework.

2.5.1 Biological Rationale and State Definitions

Analyzing SF3B1 assembly across multiple binding sites introduces direct molecular competition. Because both the functional branch point and the upstream decoy site compete for a shared pool of spliceosomal factors, complex assembly cannot be treated as two independent binding reactions. Instead, factor sequestration couples the system, such that the steady-state occupancy probability of any individual state depends on the complete transition network rather than solely on its own isolated forward and backward rates. Consequently, competitive recruitment must be formalized as a single, coupled kinetic model.

The primary objective of the model is to calculate the steady-state occupancy probabilities across all molecular configurations as a function of the sequence cis-element strengths and the pharmacological perturbation. Specifically, given a pre-mRNA sequence defined by its branch point and decoy affinities, the model determines the fractional occupancy of the following four states:

- **Free State (S_0):** The pre-mRNA sequence is unbound, with no complex assembled at either site.

- **Decoy-Bound State (S_D):** The complex is assembled at the upstream decoy site, sequestering the factor in an non-productive interaction.
- **Open BP-Bound State (S_{open}):** The complex is bound at the canonical branch point in an unstable, reversible conformation.
- **Closed BP-Bound State (S_{closed}):** The complex is stably locked at the branch point in an assembly-competent conformation ready to progress into downstream splicing stages.

Evaluating these steady-state probabilities across the drug series allows the quantification of how pharmacological perturbation shifts occupancy from functional to non-productive states across different architectural composition.

To model the competitive assembly network, standard continuous Ordinary Differential Equations (ODEs) and stochastic simulations (such as Gillespie algorithm) were considered unsuited: numerical ODE models require known rate constants or computationally expensive iterative integration over large parameter spaces, while stochastic methods track individual transient trajectories over time rather than steady-state averages.

To overcome these limitations, it has been adopted the linear framework for biochemical systems (Gunawardena, 2012). Under the assumption of memoryless continuous-time Markovian dynamics and an experimental timescale sufficiently long to ensure assembly equilibrium ($t \to \infty$). This graph approach allows the exact analytical calculation of steady-state occupancies as closed-form rational functions of the microscopic transition rates. By replacing iterative numerical ODE integrations with direct algebraic evaluation during parameter optimization, this formulation reduces computational cost while considering simultaneously the competition and conformational transitions.

2.5.2 Kinetic Assumptions and Graph Topology

The four biological states introduced above are formalized as a labeled directed graph $G = (V, E)$ ($Equation\ 1$), where the vertex set $V = \{S_0, S_D, S_{\text{open}}, S_{\text{closed}}\}$ represents the discrete microstates and each directed edge $(i \to j)$ carries a positive transition rate constant k_{ji} describing binding, unbinding, or conformational isomerization between states.

Under the *in vitro* assay conditions, SF3B1 and essential cofactors such as ATP are assumed to be well-mixed and present at constant, saturating concentrations relative to the pre-mRNA ($[\text{SF3B1}] \gg [\text{pre-mRNA}]$). This allows bimolecular binding steps and ATP-dependent reactions to be modeled with pseudo-first-order rate constants ($k_{\text{on}} = k'_{\text{on}} * [\text{SF3B1}]$), so that all

transitions remain linear and concentration-independent. The reversible binding steps are then defined by an association and a dissociation rate at each site: k_2 and k_{-2} govern binding to and release from the decoy ($S_0 \underset{k_{-2}}{\overset{k_2}{\rightleftharpoons}} S_D$), while k_1 and k_{-1} govern binding to and release from the open branch point conformation ($S_0 \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} S_{open}$).

The complex locked at the branch point can then either dissociate directly back to the free state ($S_{\text{closed}} \overset{k_4}{\rightarrow} S_0$), at rate k_4, or first form through the ATP-driven forward isomerization rate k_3, which converts the open complex into the stable closed state ($S_{\text{open}} \overset{k_3}{\rightarrow} S_{closed}$). Since the closed conformation reflects a structurally stable SF3B1:BP duplex, its dissociation rate is expected to be much smaller than the basal off-rates ($k_4 \ll k_{-1}$).

Because the initial encounter between SF3B1 and pre-mRNA depends on RNA accessibility rather than sequence specificity, the two association rates are assumed to be identical. In addition, because the assay measures binding at steady state without time courses, the model can only identify the ratio between forward and backward rates rather than their absolute values. To simplify the parameter space and remove this redundancy, both association rates were fixed to a constant value of one ($k_1 = k_2 = 1$). In this way, the estimated unbinding rates ($k_{-1}^{(0)}, k_{-2}^{(0)}$) directly reflect relative dissociation constants (K_D)

2.5.3 Sequence Affinity Modifiers and Drug Inhibition

Sequence strength is represented through residence time, that is, the duration the complex remains bound before dissociating. To capture this, two dimensionless scaling parameters were introduced, $\beta \in [0,1]$ and $\delta \in [0,1]$ that linearly modulate the microscopic off-rates: $k_{-1} = \beta \cdot k_{-1}^{(0)}$ scales the dissociation rate from the open branch point, and $k_{-2} = \delta \cdot k_{-2}^{(0)}$ scales the dissociation rate from the decoy site. Since both parameters act directly as multipliers on dissociation, strong binding sites are expected to yield lower β or δ values, reducing the effective unbinding rate and increasing residence time on the RNA, while weak binding sites are expected to yield higher values, approaching 1, accelerating factor release back into the free pool.

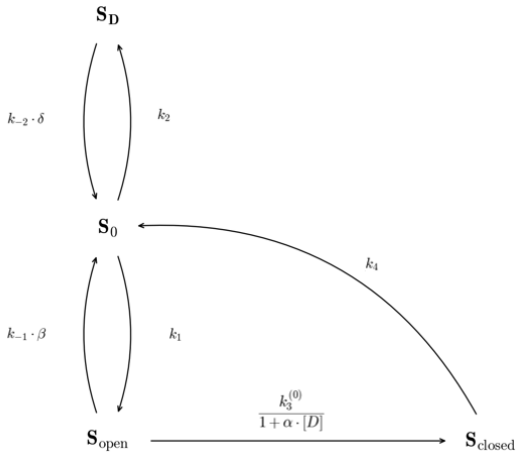
To model drug action, the structural mechanism documented for spliceostatins was incorporated (Cretu et al., 2016). These compounds bind SF3B1 and selectively block the open-to-closed structural transition without directly

preventing initial pre-mRNA recruitment. This inhibition was modeled by introducing a hyperbolic inhibition term into the rate k_3:

$$k_3([D]) = \frac{k_3^{(0)}}{1 + \alpha[D]}$$

where $[D]$ denotes the experimental drug concentration, $\alpha \geq 0$ is the global drug sensitivity parameter, representing the apparent affinity ($\frac{1}{K_I}$) of the inhibitor for the open complex, and $k_3^{(0)}$ is the baseline rate in the absence of drug. This hyperbolic formulation captures saturable pharmacological inhibition: at $[D] = 0$, the transition proceeds at its basal rate $k_3^{(0)}$while increasing drug doses progressively suppress the rate of transition into the stable, closed conformation (S_{closed}). Because the inhibitor targets the core SF3B1 complex rather than the RNA sequence directly, α is treated as a global shared parameter, invariant across all sequence architectures.

The overall system representation is than represented as:



Equation 1

2.5.4 Steady-State Solution via the Matrix-Tree Theorem

Let $p_i(t)$ denote the probability of the system occupying state S_i at time t. The temporal evolution of the state vector $p(t) = \left(p_0(t), p_D(t), p_{\text{open}}(t), p_{\text{closed}}(t)\right)^T$ is governed by the continuous-time Markov master equation:

$$\frac{dp(t)}{dt} = L(G)\, p(t)$$

where $L(G)$ is the graph Laplacian matrix. Off-diagonal entries $L_{ji} = k_{ji}$ $(i \neq j)$ correspond to the microscopic rate from state S_i to S_j, while diagonal elements $L_{ii} = -\sum_{j \neq i} k_{ji}$ ensure the conservation of total probability:

$$L(G) = \begin{pmatrix} -(k_2 + k_1) & \delta * k_{-2}^{(0)} & \beta * k_{-1}^{(0)} & k_4 \\ k_2 & -\delta * k_{-2}^{(0)} & 0 & 0 \\ k_1 & 0 & -\left(\beta * k_{-1}^{(0)} + k_3([D])\right) & 0 \\ 0 & 0 & k_3([D]) & -k_4 \end{pmatrix}$$

At steady state ($\frac{dp}{dt} = 0$), the system can be reduced to $L(G)\, p^* = 0$ subject to $\sum_i p_i^* = 1$. Because the four-state network is strongly connected without absorbing sinks, the null space $\ker(L(G))$ is one-dimensional, this property guarantee a unique, positive steady-state distribution (Gunawardena, 2012).

Rather than integrating differential equations numerically, exact algebraic solutions are obtained via the Matrix-Tree Theorem. For each state S_i, the basis polynomial $\rho_i(G)$ of $\ker(L(G))$ is computed by summing the products of edge rates over all directed spanning trees rooted at S_i:

$$\rho_i(G) = \sum_{T \in \Theta_i(G)} \left(\prod_{(u \to v) \in E(T)} k_{vu} \right)$$

Because the four-state topology contains a single directed spanning tree rooted at each vertex, the tree polynomials are derived directly as:

$$\rho_0(G) = (\delta \cdot k_{-2}^{(0)}) \cdot (\beta \cdot k_{-1}^{(0)}) \cdot k_4$$

$$\rho_D(G) = k_2 \cdot (\beta \cdot k_{-1}^{(0)}) \cdot k_4$$

$$\rho_{\text{open}}(G) = (\delta \cdot k_{-2}^{(0)}) \cdot k_1 \cdot k_4$$

$$\rho_{\text{closed}}(G) = (\delta \cdot k_{-2}^{(0)}) \cdot k_1 \cdot k_3([D])$$

Normalizing each polynomial by the total partition sum allows to compute the exact steady-state occupancy probabilities:

$$p_i^* = \frac{\rho_i(G)}{\rho_0(G) + \rho_D(G) + \rho_{\text{open}}(G) + \rho_{\text{closed}}(G)}$$

This closed-form rational formulation calculates state occupancies directly as a function of sequence affinity scalers (β, δ) and drug concentration ($[D]$) without iterative numerical integration.

2.6 Computational Implementation and Reproducibility

The entire modeling, simulation, and inference pipeline was implemented in *Python 3.12*. Data structures and numerical operations relied on *pandas* (data handling) and *numpy* (array computation); the kinetic graph and its steady-state solution were built with *networkx* (directed graph representation) while parameter optimization used *scipy.optimize* (L-BFGS-B). Statistical comparisons (e.g. Mann-Whitney U tests) used *scipy.stats*. Figures were produced with *matplotlib* and *seaborn*.

The compositional normalization pipeline (CLR transformation, Kneedle elbow detection, Leave-One-Condition-Out validation, cross-drug intersection) was implemented separately in R (v4.4.2), using *tidyverse* for data manipulation, *zCompositions* for compositional data handling.

The code was organized into dedicated modules, each responsible for one stage of the pipeline:

- **model_database.py** : catalogue of candidate kinetic models: for each model, defines which rate constant(s) are drug-modulated and through which functional form (e.g. Hill decay, linear decrease/increase, Hill activation).
- **simulation_core.py** : builds the directed graph for a given model from model_database, and computes steady-state occupancy probabilities via the Matrix-Tree Theorem.
- **parameter_function.py** : generates and constrains candidate parameter sets under the biological assumptions described in Methods (e.g. comparable BP/Decoy association rates, higher Decoy off-rate relative to BP), used both for simulation and as initialization for the multistart optimization.
- **fitting_core.py** : core fitting optimization computes observed $\log_2$FC from the data, evaluates model-predicted $\log_2$FC via *_p_bound*, and implements the weighted least-squares objective minimized by *scipy.optimize.minimize*.

CLR normalization pipeline (R) implements the reference-based Compositional Data Analysis workflow described in Methods.

To avoid numerical issues with the estimate of kinetic parameters spanning several orders of magnitude, shared rate constants were optimized in base_{10} logarithmic space, while condition-specific scaling parameters (β, δ) were bounded directly in linear space within [0,1]. Multi-start initialization vectors and bootstrap resampling were generated using *numpy.random.default_rng* with fixed random seeds, ensuring full computational reproducibility of both the point-estimate fit and the bootstrap uncertainty analysis.

2.7 Parameter Estimation and Statistical Inference

Having defined a four-state graph and derived its analytical steady-state probabilities, the next step consists in parameter estimation to evaluate whether the model captures the experimental behavior under pharmacological stress. Beyond minimizing the fitting error, the optimization must verify if the inferred parameters capture biological mechanisms rather than curve-fitting artifacts generated by an optimizer.

Scope and Evaluation Strategy

In this work, the focus is on estimating the parameters, checking their identifiability, and quantifying their uncertainty within the chosen model structure, rather than on evaluating the model's ability to generalize to new, unseen data. Once that this fundamental aspects are verified, it can be tested the predictive capacity and the overfitting of the model by splitting the data into training, used to estimate the parameters, and test set used to compute the accuracy metrics.

2.7.1 Objective Function and Joint Optimization

The experimental library is partitioned into distinct architectural conditions defined by the combination of branch point and decoy binding strengths. Within each category, read counts naturally exhibit variance: the decoy has seven discrete sequence designs, while continuous branch point scores are discretized into strength classes.

The parameter space is split into two complementary groups: the *Global Shared Parameters* ($\Phi = \left\{k_{-1}^{(0)}, k_{-2}^{(0)}, k_3^{(0)}, k_4, \alpha\right\}$), which are the kinetic rate constants and the drug-inhibition constant estimated by the model (with k_1 and k_2 fixed to 1 as described in Section 2.5.2); and the *Condition-Specific Scaling Parameters* ($\Theta_c = \{\beta_c, \delta_c\}$), which scale the baseline dissociation rates ($k_{-1} = \beta_c \cdot k_{-1}^{(0)}$ and $k_{-2} = \delta_c \cdot k_{-2}^{(0)}$) to account for the affinity differences of each architectural condition c.

A joint optimization was implemented where all experimental conditions are fitted simultaneously within a single objective function, rather than fitting each architectural subset independently or estimating a single fully global parameter vector: the first approach would let shared parameters such as α drift across conditions, while the second wouldn't consider the differential responses between sequence classes that this model is meant to capture. Joint optimization ensures shared kinetic constraints across the entire dataset while isolating condition-specific affinity differences within the scaling factors Θ_c.

To test the kinetic framework across the clearest biological contrasts without introducing noise from intermediate groups, the joint optimization was focused specifically on the four extreme boundary conditions defined in Section 3.1.1 (combinations of Strong or Weak BP with Absent or Strong Decoy, all within the Strong PYT background). These four cases represent the full range of system behaviors from maximal drug sensitivity to complete baseline insensitivity, and from normal branch point assembly to decoy trapping. Focusing on these extreme cases provides a direct and controlled benchmark to test whether the model parameters can be identified.

The model parameters are estimated by minimizing a weighted mean squared error (WMSE) between the experimentally observed and model-predicted $\log_2$FC, summed across all architectural conditions and drug concentration points:

$$\mathcal{L}(\boldsymbol{\theta}) = \sum_c \sum_i w_{c,i} \left(\mathrm{LFC}_{\mathrm{obs},c,i} - \mathrm{LFC}_{\mathrm{pred},c,i}(\boldsymbol{\theta})\right)^2$$

where c indexes the experimental sequence condition, i indexes the drug concentration, and where $w_{c,i} = 1/\sigma_{c,i}^2$ is the inverse-variance weight. Here, $\sigma_{c,i}^2$ is the variance calculated across all individual sequences in condition c at drug dose i. Since replicate experiments aren't available, this variance measures how much sequences in the same group differ from one another at

that dose. This gives less weight to noisy data points without hiding changes caused by the drug.

Because the gel selection isolates all RNA sequences bound to the SF3B1 complex regardless of binding site or conformation, the total predicted bound fraction corresponds to the sum of all bound microstate probabilities:

$$P_{\text{bound}}(c, i; \boldsymbol{\theta}) = p_D^*(c, i) + p_{\text{open}}^*(c, i) + p_{\text{closed}}^*(c, i)$$

The model-predicted $\log_2$FC at drug concentration i relative to the untreated control condition (i=0) is therefore given by:

$$\text{LFC}_{\text{pred},c,i}(\boldsymbol{\theta}) = \log_2\left(\frac{P_{\text{bound}}(c, i; \boldsymbol{\theta})}{P_{\text{bound}}(c, 0; \boldsymbol{\theta})}\right)$$

The complete parameter vector θ contains both the shared parameters ($\Phi = \left\{k_{-1}^{(0)}, k_{-2}^{(0)}, k_3^{(0)}, k_4, \alpha\right\}$) and the condition-specific scalers ($\Theta_c = \{\beta_c, \delta_c\}$ for each condition c).

To maintain numerical stability across several orders of magnitude, shared kinetic rate constants were optimized in $\log_{10}$ space within bounded domains of $[-5,5]$. This range was selected as a pragmatic compromise between search-space dispersion and computational cost; narrower bounds frequently led to parameter estimates clustering against optimization boundaries, signaling the need for a wider dynamical interval. The drug sensitivity parameter α was bounded within [0,100] to match the normalized experimental drug titration scale (0-100). Conversely, the condition-specific scalers β and δ were strictly constrained to the unit interval [0,1]. Rather than estimating physical values, these parameters should capture relative site strength: values closer to 0 should indicate stronger binding with longer residence times, while values closer to 1 should represent weaker, more unstable interactions.

The loss function was minimized using the bounded L-BFGS-B algorithm (Byrd et al., 1995) implemented in *scipy.optimize.minimize*.

2.7.2 Uncertainty Quantification

The standard fitting procedure results in a single point estimate by minimizing the loss against the empirical mean $\log_2$ FC of each condition. While this mean

serves as a proxy for average group behavior, a single estimate provides no information about parameter uncertainty. Without an estimate of variance, it is impossible to evaluate parameter identifiability, assess fit stability, or statistically determine if differences between fitted condition-specific scalers are meaningful.

To quantify parameter uncertainty without relying on parametric distributional assumptions, a non-parametric bootstrap approach (Efron & Tibshirani, 1993) was implemented. This strategy was chosen over fitting each sequence independently: per-sequence optimization would capture sequence-to-sequence biological heterogeneity rather than the standard error of the aggregate parameter estimates.

The bootstrap framework specifically quantifies the sampling variance of the joint estimate, providing empirical confidence intervals for both shared and condition-specific parameters.

At each bootstrap iteration $b \in \{1, \dots, B\}$, the original subset of N_c sequences ($N_c \approx 4{,}000$) within each condition c is resampled with replacement to construct a pseudo-replicate sample of the exact same size N_c. Resampling is performed independently within each architectural condition rather than on the whole library, ensuring that every condition preserves its original sample size and architectural composition across iterations. On the resampled dataset, the mean $\log_2$ FC, standard deviation $\sigma_{c,i}$, and inverse-variance weights ($w_{c,i} = 1/\sigma_{c,i}^2$) are recomputed for each drug concentration point i, directly propagating the sampling variability of the sequences into the weights and target values passed to the objective function. These resampled summary metrics across all conditions are then passed to the optimization problem jointly, since fitting all conditions together is necessary because the global kinetic parameters (Φ) are shared across the entire system and cannot be estimated from single isolated subsets.

Because the objective function is non-linear and its loss landscape is non-convex, a standard gradient descent run from a single initial guess can converge to a suboptimal local minimum rather than the global one. To minimize this problem, at each bootstrap iteration the optimization is repeated from 35 independent, randomly generated initial parameter vectors, and only the parameter set with the lowest loss value is retained for that step. The number of random starts was evaluated empirically by increasing initialization points until the distribution of minimized losses no longer exhibited multimodal loss clusters. The final uncertainty analysis was executed across B=150 bootstrap iterations using 35 random restarts per iteration, generating an empirical parameter distribution $\{\hat{\boldsymbol{\theta}}^{(1)}, \dots, \hat{\boldsymbol{\theta}}^{(B)}\}$ to compute confidence intervals and evaluate parameter robustness.

Across the $B = 150$ bootstrap runs, 150 optimal values are obtained for each parameter which are sorted in ascending order. The final estimate is reported as the median (the 50th percentile). The 95% confidence interval is determined directly from the empirical distribution using the percentile method: the lower bound is the 2.5th percentile and the upper bound is the 97.5th percentile.

2.7.3 Goodness-of-Fit Metrics

To evaluate model calibration, two complementary metrics were computed: the global coefficient of determination (R^2) and the condition-specific Root Mean Square Error (RMSE).

The global R^2 evaluates the overall proportion of variance in the observed $\log_2$ FC explained by the joint optimization across all conditions and drug concentrations:

$$R^2_{w,global} = 1 - \frac{\sum_c \sum_i w_{c,i} \left(LFC_{obs,c,i} - LFC_{pred,c,i}\right)^2}{\sum_c \sum_i w_{c,i} \left(LFC_{obs,c,i} - \overline{LFC}_{w,global}\right)^2}$$

where c indicates the sequence condition, i is the drug concentration, and $\overline{\mathrm{LFC}}_{\mathrm{global}}$ is the grand mean across all data points. The weight $w_{c,i} = 1/\sigma^2_{c,i}$ is the inverse variance for each point, ensuring that the metric directly accounts for measurement noise just like the objective function during optimization.

While the global R^2 confirms whether the model captures the broad dynamic range across the whole library, computing a local R^2 within a single condition can be mathematically unstable: for biologically flat conditions with minimal drug response (ΔLFC ≈ 0), the total sum of squares approaches zero ($SS_{tot} \rightarrow 0$), causing R^2 to produce large negative values even when residuals are near zero. To avoid this artifact, intra-condition accuracy is instead quantified using the absolute RMSE:

$$\mathrm{RMSE}_c = \sqrt{\frac{1}{N_d} \sum_{i=1}^{N_d} \left(\mathrm{LFC}_{\mathrm{obs},c,i} - \mathrm{LFC}_{\mathrm{pred},c,i}\right)^2}$$

which retains the same physical scale as the observed data ($\log_2$ FC) and measures the standard deviation of residuals. This measure is evaluated against the Experimental Range Length of that condition,

$$\Delta\mathrm{LFC}_c = \max_i\left(\mathrm{LFC}_{\mathrm{obs},c,i}\right) - \min_i\left(\mathrm{LFC}_{\mathrm{obs},c,i}\right)$$

Comparing the RMSE to the Experimental Range Length ($\Delta\mathrm{LFC}_c$) allows to quantify the error relative to the actual biological signal of each curve. An RMSE of 0.1 represents an excellent fit for a responsive condition that varies across 2.0 units of LFC, but it represents significant error for a flat condition that only varies across 0.1 units. Normalizing the local error by the dynamic range of that condition confirms whether the model truly traces the observed trajectory rather than merely predicting a flat line through the data.

2.7.4 Parameter Validation and Cross-Drug Consistency

To evaluate the biological robustness and stability of the model, the parameter estimation was executed independently on the two pharmacological datasets, PladB and H3B.

Because PladB and H3B are distinct chemical entities tested in separate experiments, comparing the two fits allows for testing whether the optimization converges to real signals rather than artifacts. Specifically, the shared kinetic parameters Φ describe the core SF3B1 machinery before drug exposure, whereas the condition-specific scalers (β and δ) capture the intrinsic binding properties of the RNA constructs. Because both assays employ the same cellular extracts and the identical mutagenesis library, neither set of parameters should depend on which compound was titrated, meaning their fitted values are expected to be consistent between PladB and H3B.

For each parameter θ, the independent non-parametric bootstrap approach result in two empirical sampling distributions, $\mathcal{D}_{\mathrm{PladB}}(\theta)$ and $\mathcal{D}_{\mathrm{H3B}}(\theta)$. Their consistency was evaluated using the non-parametric two-sided Mann-Whitney U test. A non-parametric test was chosen because the bootstrap parameter distributions often deviate from normality, displaying skewness, heavy tails, or boundary-stacking behavior that violates the distributional assumptions of standard parametric tests such as Student's t-test. where a non-significant result ($p \geq 0.05$, denoted ns) confirms that the parameter distributions are

statistically indistinguishable between the two independent experimental series.

The estimated parameters are also evaluated against the biophysical expectations of SF3B1 assembly. The basal dissociation rate from the decoy site is expected to be higher than from the open branch point ($k_{-2}^{(0)} > k_{-1}^{(0)}$), reflecting that non-functional decoy sites form less stable baseline contacts; the forward transition locking the complex into the closed state ($k_3^{(0)}$) is expected to be faster than basal unbinding ($k_3^{(0)} \gg k_{-1}^{(0)}$), consistent with an ATP-dependent commitment step; the dissociation rate from the closed conformation (k_4) is expected to be near zero ($k_4 \ll k_{-1}^{(0)}$), capturing the stability of the clamped spliceosome complex.

Lastly, the fitted condition-specific parameters are expected to follow the affinity hierarchy ($\beta_{Strong} < \beta_{Intermediate} < \beta_{Weak}$ for the branch point, and $\delta_{Strong} < \delta_{Intermediate} < \delta_{Weak}$ for the decoy), confirming that lower numerical multipliers correspond to longer residence times on strong cis-regulatory elements.

3 Results and Discussion

3.1 Preliminary Exploratory Analysis

3.1.1 Dose-Response Landscape Across Architectural Features

To characterize how the architectural components of pre-mRNA sequences modulate sensitivity to spliceosome inhibitors, the dose-response behavior across all library design combinations were evaluated. For each sequence, the rate of change across the drug titration was quantified by fitting a linear regression model:

$$\mathrm{LFC}_i = \beta \cdot [D]_i + \epsilon_i$$

where the slope (β) represents the dose sensitivity relative to the untreated control. The median slopes were stratified across all combinations of polypyrimidine tract strength (PYT), branch point strength class (BP_class), and decoy site strength where 7 is decoy absence and than the strength increases (DECOY) to generate an architectural sensitivity landscape.

Figures 3 and 4 display the dose-response sensitivity landscape across all structural combinations, split by weak (Fig. 3) and strong (Fig. 4) PYT backgrounds, respectively. In each heatmap, the horizontal axis stratifies sequences across decoy site strength bins (ranging from decoy absence to increasing decoy affinity), while the vertical axis partitions them by branch point affinity class (Strong, Intermediate, Weak). Individual cells report the median linear regression slope (β), accompanied by a diverging color gradient where negative values indicate drug-induced complex displacement and positive values denote complex accumulation or retention.

Impact of PYT Strength on Drug Sensitivity

The landscape reveals an evident contrast driven by the polypyrimidine tract:

- **Weak PYT (Flat and Homogeneous Landscape):** Under a weak PYT context (Fig. 3), the response across the treatment is largely uniform and insensitive to branch point quality. Across all decoy columns, the different BP classes display virtually identical slopes ($\beta \approx 0.05$ to 0.15), indicating that without a functional PYT to stabilize U2AF recruitment, variations in BP sequence do not modulate drug response. Also, all slopes remain slightly positive, indicating that these sequences exhibit a modest increase in relative complex retention under drug compared to control. The slight non-monotonicity observed across intermediate decoy columns (e.g., column means of 0.05 for No Decoy, 0.14 for Weak

Decoy, and 0.08 for Strong Decoy) likely still reflects sequencing compositional artifacts.

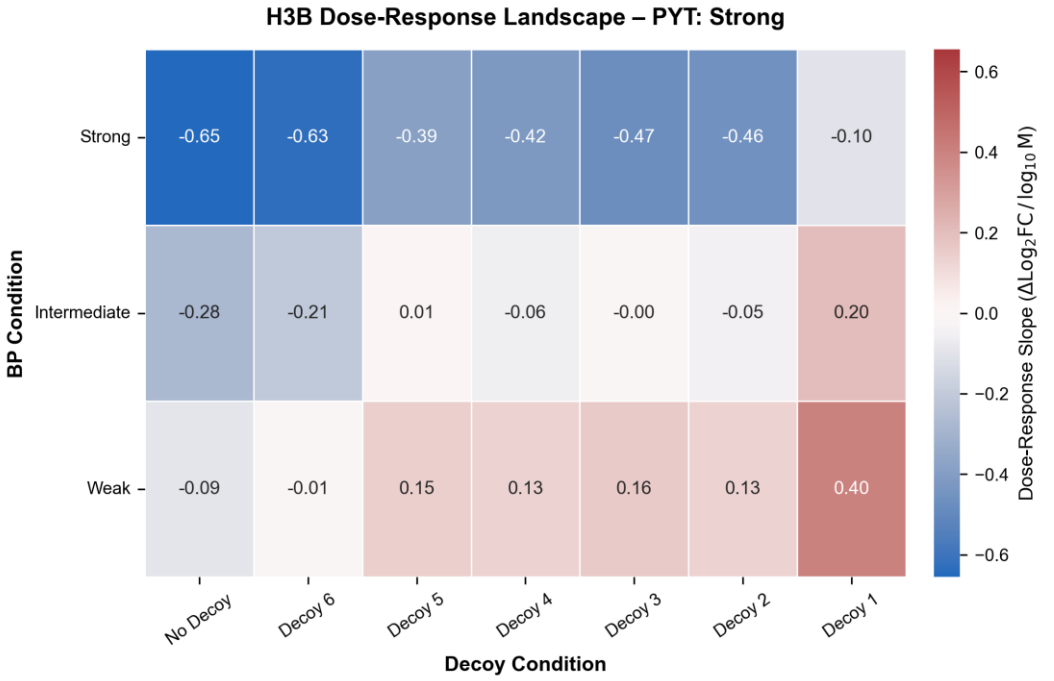


Figure 3 **| Dose-response sensitivity landscape under a weak Polypyrimidine Tract (PYT) background.** Heatmap showing the median dose-response slopes (β) calculated across drug titration for all combinations of Branch Point (BP) classes (vertical axis) and decoy site strength (horizontal axis; ranging from condition 7 [absence of decoy] to increasing decoy strength). Values near zero or slightly positive across all bins reflect a flat landscape where branch point and decoy variations exert minimal regulatory influence in the absence of a strong PYT.

- **Strong PYT (Structured Landscape):** In contrast, the strong PYT background (Fig. 4) exhibits a structured, dynamic-range landscape governed by the interaction between the branch point and the decoy site.

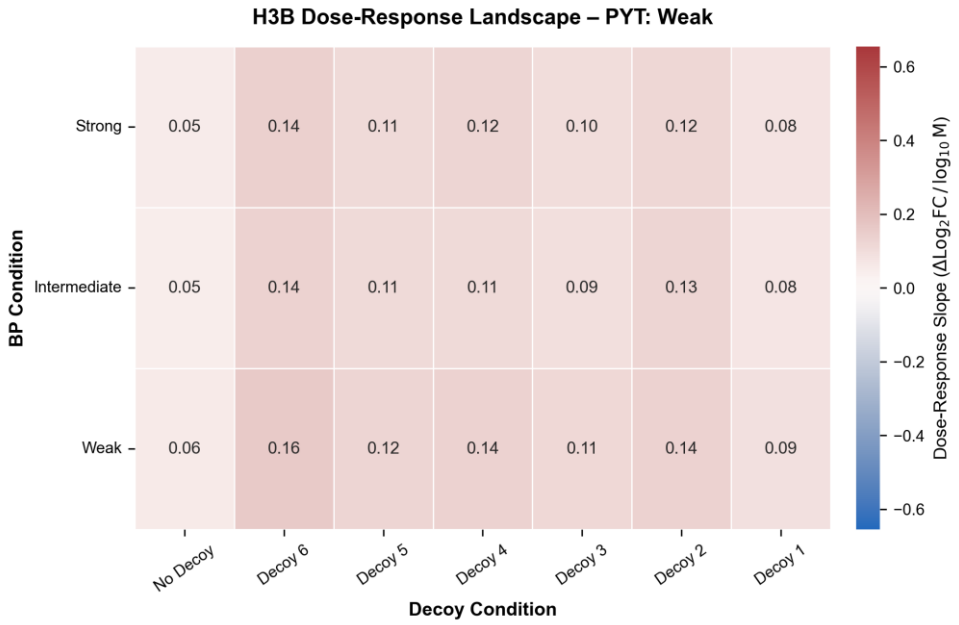


***Figure 4* | Dose-response sensitivity landscape under a strong Polypyrimidine Tract (PYT) background.** Heatmap illustrating the median dose-response slopes (β) across BP affinity classes (vertical axis) and decoy site strength conditions (horizontal axis). The diverging color scale highlights the dynamic interplay between cis-elements: canonical BP repression depth scales with branch point strength (vertical gradient), whereas decoy strength progressively rescues complex retention, shifting slopes toward positive values (horizontal gradient).

Cis-Element influence Under Strong PYT

Within the strong PYT landscape, two trends emerge:

1. **Branch Point Strength Modulates Repression Depth:** Along any given decoy column, moving from Strong to Weak BP systematically shifts the slopes towards less negative or more positive values. In the absence of a decoy, Strong BP sequences are severely repressed by the drug ($\beta = -0.65$), Intermediate BPs show moderate reduction ($\beta = -0.28$), and Weak BPs show minimal change ($\beta = -0.09$). This is biologically consistent with the drug mechanism: in control conditions, Strong BPs drive efficient spliceosome assembly and thus suffer the largest absolute drop upon pharmacological inhibition. Weak BPs, having low baseline occupancy in the control, show a smaller range of repression.

2. **Decoy Strength Drives Alternative Site Occupancies:** Along any given BP row, increasing the decoy strength consistently increases the slope from negative to positive values. In the presence of strong decoys, pharmacological inhibition of the canonical BP force SF3B1 complexes to assemble at the decoy site. Because spliceostatins impair proofreading without abolishing initial binding, complexes become trapped at the high-affinity decoy site, resulting in positive slopes ($\beta >$

0) where sequences are more bound under drug treatment than under basal conditions.

Selection of the Four Extreme Boundary Conditions

The clear dynamic observed in the strong PYT regime established the four extreme boundary conditions used to reduce model complexity while preserving the biology:

- **Strong BP / Decoy Absent:** Represents purely canonical, highly efficient assembly that suffers maximal drug inhibition.
- **Strong BP / Strong Decoy:** Represents the high-competition regime between canonical assembly and decoy trapping.
- **Weak BP / Decoy Absent:** Represents a low-affinity canonical baseline with minimal assembly turnover.
- **Weak BP / Strong Decoy:** Represents maximal redirection of the complex toward the decoy site under pharmacological stress.

Cross-Compound Concordance

Applying identical linear regression and stratification to the Pladienolide B and H3B-8800 datasets produced consistent slope landscapes across all architectural regimes. This concordance demonstrates that the observed sensitivity shifts are driven by the properties of the sequence architectures rather than drug related effects. As a consequence these boundary conditions provide an accurate reference where to inspect the four-state kinetic model.

3.1.2 Differential Assembly Across Extreme Regimes

Figure 5 illustrates the distribution of $\log_2$ Fold Change (LFC, computed on CLR-normalized read counts relative to the untreated control) across drug titration points for both H3B-8800 (Fig. 5a, left) and Pladienolide B (Fig. 5b, right). In each panel, the response is stratified across four subplots representing the extreme architectural combinations of branch point quality (BP_class) and decoy site affinity under the Strong PYT background.

Each subplot displays violin distributions across escalating drug concentrations (x-axis) against LFC values (y-axis), with internal medians tracking dosage sensitivity.

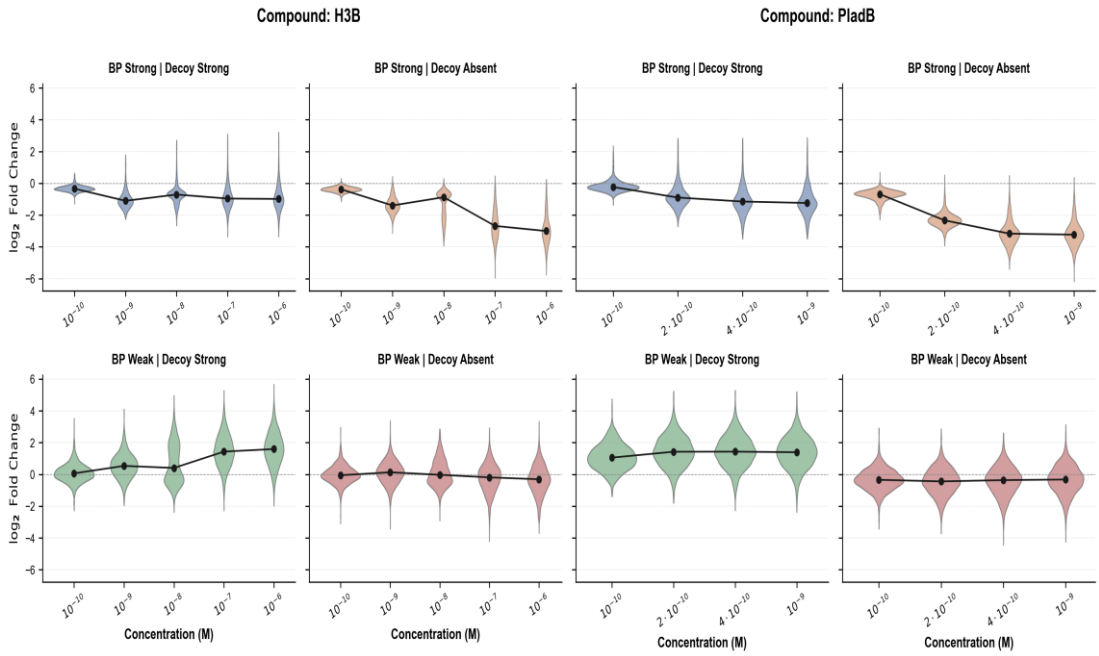


***Figure 5*| Dose-dependent response profiles across extreme architectural regimes under SF3B1 inhibition.** Violin plots showing the distribution of log_2 Fold Change (LFC) calculated on CLR-normalized read counts relative to the untreated control across drug titration points for (a) H3B-8800 and (b) Pladienolide B. In both panels, rows or facets partition the four extreme structural combinations within the Strong PYT background: Strong BP / No Decoy, Strong BP / Strong Decoy, Weak BP / No Decoy, and Weak BP / Strong Decoy. Horizontal dashed lines mark the baseline (LFC = 0). The plots capture the concordance between compounds, showing progressive canonical repression (Strong BP / No Decoy), decoy-mediated buffering (Strong BP / Strong Decoy), baseline insensitivity (Weak BP / No Decoy), and aberrant complex accumulation (Weak BP / Strong Decoy), alongside broadening variance at higher drug dosages.

Cross-Drug Concordance and General Trends

The median response profiles across the four extreme architectures show consistent behavior between the two distinct compounds (Fig. 5a vs Fig. 5b):

- **Strong BP / No Decoy:** Shows the largest progressive decrease in $\log_2$ FC as drug concentration increases.
- **Strong BP / Strong Decoy:** Exhibits a monotonic decrease with drug titration, though the magnitude of the drop is dampened compared to the No Decoy condition.
- **Weak BP / No Decoy:** Remains essentially flat across all drug concentrations ($\log_2$ FC ≈ 0).
- **Weak BP / Strong Decoy:** Displays a progressive increase in $\log_2$ FC above baseline with increasing drug concentration.

Beyond median shifts, the violin reveal an increase in intra-condition variance under pharmacological perturbation, particularly in the Strong BP conditions. The distributions changes from narrow, unimodal shapes at low concentrations to broader profiles at higher doses, indicating residual heterogeneity within each condtion.

Biophysical Interpretation of the Response Regimes

These distinct trajectories align with the proposed mechanism of SF3B1 pharmacological inhibition:

- **Maximal Repression (Strong BP / No Decoy):** In the basal state, strong canonical branch points recruit SF3B1 with high efficiency. Because spliceostatins block the transition into the stable closed state and no alternative binding site is available, complex assembly is reduced, resulting in a large negative $\log_2$ FC.
- **Decoy-Mediated Buffering (Strong BP / Strong Decoy):** When a high-affinity decoy site is present alongside a strong BP, drug-induced displacement from the canonical branch point allows complexes to be captured upstream. This alternative recruitment partially compensate the loss of canonical binding, yielding a lower decrease in total bound reads.
- **Insensitive Baseline (Weak BP / No Decoy):** Sequences lacking both an efficient branch point and an upstream decoy exhibit low baseline binding in the untreated control. Consequently, pharmacological disruption produces less changes across all drug points, serving as an internal drug-insensitive negative control.
- **Aberrant Trapping (Weak BP / Strong Decoy):** In the control condition, weak branch points show very low baseline binding. When the drug is added, it inhibits the canonical branch point, redirecting free SF3B1 complexes toward the high-affinity decoy site. Because the drug blocks proofreading and factor release without stopping initial binding, the complex becomes trapped at the decoy. This accumulation results in a net increase in total bound sequences compared to the untreated control ($\log_2 \text{FC} > 0$).

Summary

These results confirm that the CLR-normalized enrichment scores capture distinct, biologically consistent responses driven by the interaction of cis-regulatory strength and pharmacological stress. While the assay readout cannot resolve binding position directly, comparing these extreme boundary architectures can be a proxy to the competitive expected.

3.2 Compositional Normalization

To demonstrate the efficacy of Compositional Data Analysis (CoDA) framework has been performed a direct comparison between the standard normalization (naive total-count library scaling, as described in Section 2.1.5)

and the Centered Log-Ratio (CLR) correction.

3.2.1 The Reference Class: Correcting Composition

The necessity of a reference-based CLR normalization over standard library-size scaling is demonstrated by analyzing the negative control class (PYT_Weak / BP_Weak / No Decoy). By design, this architecture lacks both a functional branch point and an upstream decoy site, meaning it should not recruit SF3B1 efficiently and should remain biologically unresponsive to drug treatment.

As shown in Figure 6, the plots trace the normalized signal (y-axis) across increasing drug doses (x-axis) for both treatments, comparing naive total-count scaling directly to reference-based CLR. Naive normalization creates a false trend: as strong-binding sequences drop out during drug titration, weak sequences look like they are increasing. In addition, the naive baseline starts at around -3, which distorts the real scale. In contrast, the reference-based

CLR method removes this read-redistribution artifact, keeping the signal flat near zero across all drug doses, as expected for an inactive control.

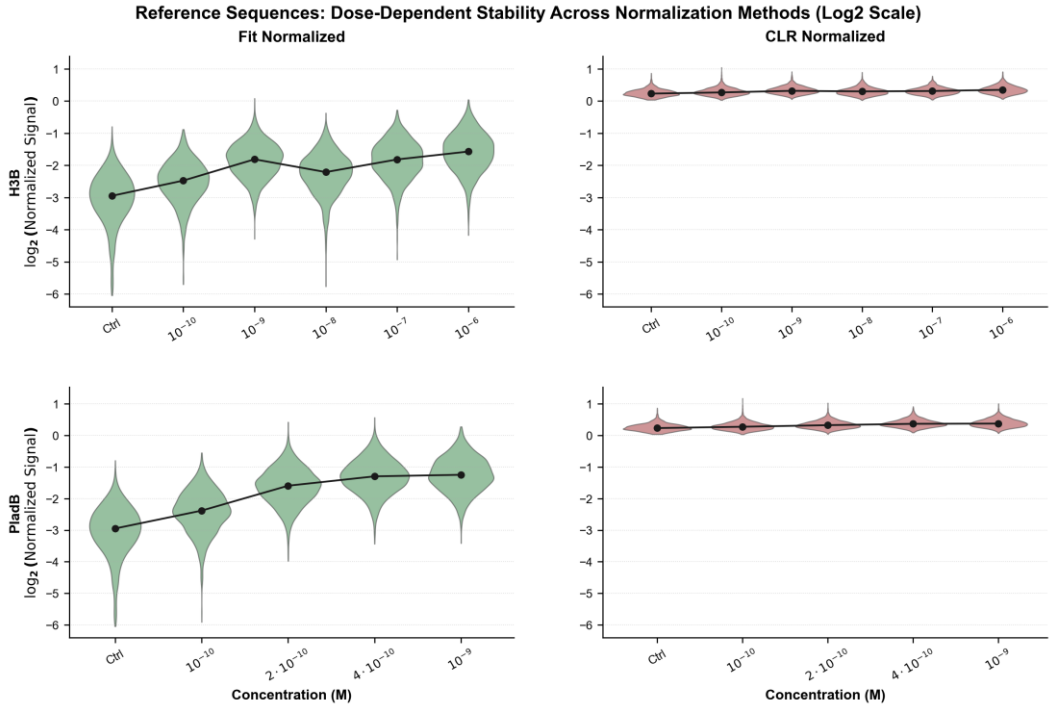


Figure 6 | **Comparison between reference-based CLR and naive total-count normalization on negative control sequences.** Signal response of the negative control class (PYT_Weak / BP_Weak / No Decoy) across increasing concentrations of H3B-8800 and Pladienolide B. Curves compare naive total-count scaling against reference-based CLR. Naive normalization shows an artificial increase caused by read redistribution after strong binders drop out. Reference-based CLR keeps the values close to zero (LFC ≈ 0), correctly showing that these sequences do not respond to the drugs.

3.2.2 Global Stratification by Polypyrimidine Tract and Branch Point Strength

The impact of the normalization method is consistent across the entire dataset. Figures 7 and 8 show the data grouped by Polypyrimidine Tract (PYT) and Branch Point (BP) strength under both drug treatments, comparing naive total-count scaling (left panels) to reference-based CLR (right panels). The x-axis shows the increasing drug doses, while the y-axis shows the normalized response. This grouping helps check whether each method matches known biology:

- **PYT Weak Dynamics** (Fig. 7) : A weak PYT strongly impairs initial SF3B1 assembly. Since the drugs should not promote assembly here, these sequences are expected to remain flat across all drug doses, regardless

of BP strength. With naive normalization (left panels), all BP classes show a false increase across both drugs. Applying CLR (right panels) flattens these curves, confirming that the apparent increase was an artifact of read redistribution.

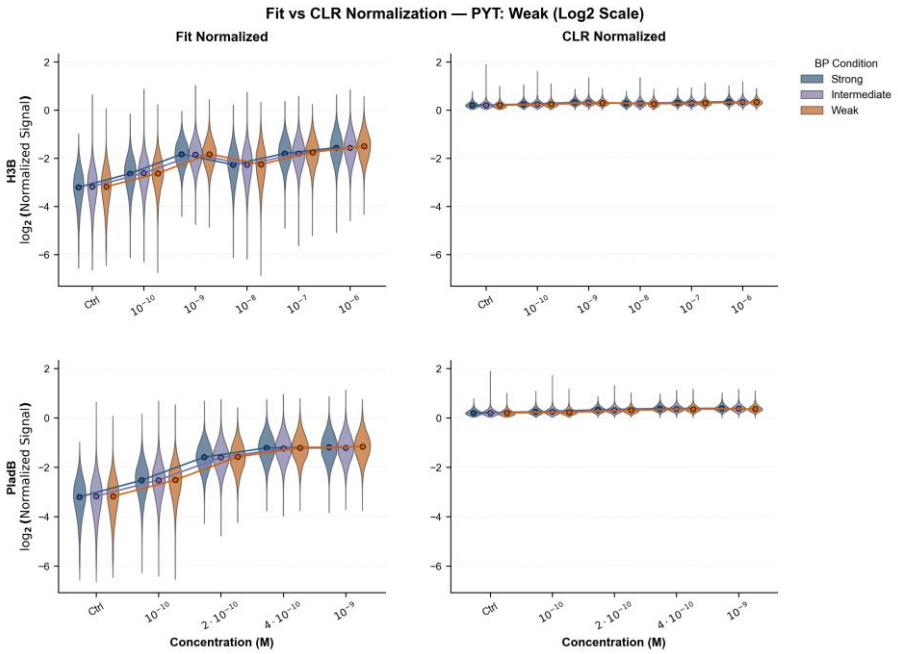


Figure 7 **| Comparison of normalization methods across Branch Point classes under a weak PYT context.** Normalized dose-response profiles for weak PYT sequences across increasing concentrations of H3B-8800 and Pladienolide B, grouped by Branch Point strength. Left panels display naive total-count normalization, which shows an artificial upward trend across all classes. Right panels display reference-based CLR normalization, which keeps trajectories flat around zero as biologically expected for non-functional transcripts.

- **PYT Strong Dynamics** (Fig. 8) : With a strong PYT, the response depends mainly on the BP. Weak BP sequences have minimal baseline binding and should stay flat. Functional sequences (BP_Intermediate and BP_Strong) should show a dose-dependent drop, with Strong BP falling the most. Under naive normalization (left panels), weak BP sequences show an artificial increase. The CLR normalization (right panels)

corrects this: weak BP sequences remain flat, while intermediate and strong BP sequences follow the expected decrease.

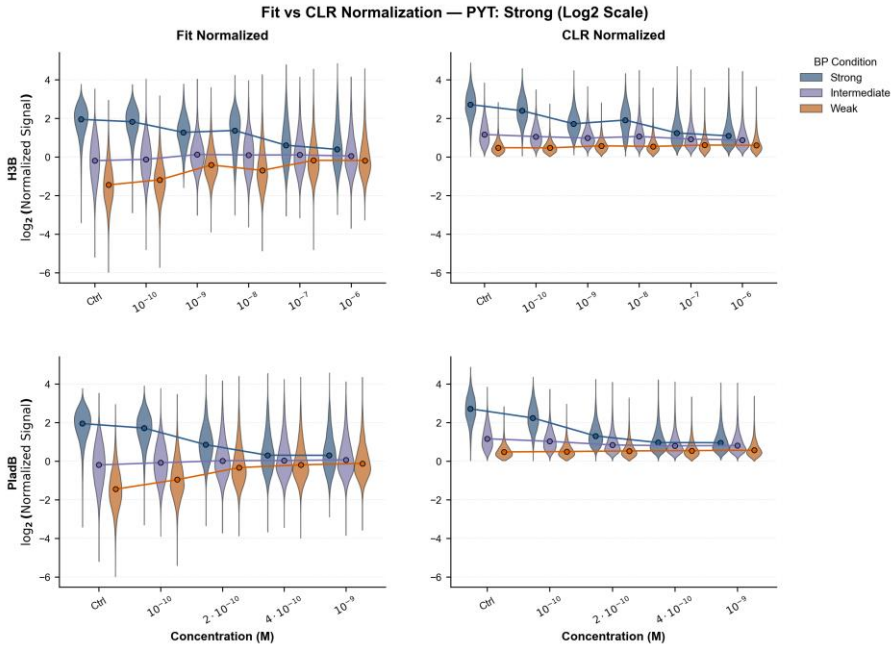


Figure 8 **| Comparison of normalization methods across Branch Point classes under a strong PYT context.** Normalized dose-response trajectories for strong PYT sequences across drug titration points, stratified by Branch Point strength (Strong, Intermediate, Weak). Left panels show naive total-count normalization, where weak BP classes artificially increase. Right panels show reference-based CLR normalization, correctly keeping weak BPs flat while preserving the expected dose-dependent drop for intermediate and strong BP classes.

This comparison proves that without applying the CLR framework, are introduced some technical artifacts incoherent with the biological expected trends. Only by correcting them with the proper computational approach will lead to fit the desined model with more biological coherent data.

3.3 Model Calibration and Identifiability

Before inspecting parameter estimates and their biological meaning, the model was calibrated on the four extreme sequence architectures identified in the preliminary analysis (Strong and Weak BP crossed with Absent and Strong Decoy, in the Strong PYT background). Using these boundary conditions allows the optimization to be tested against the widest response range in the dataset while avoiding the high internal variability of intermediate sequence classes.

3.3.1 Practical Identifiability via Multistart Analysis

Before evaluating the kinetic parameters, it is essential to ensure that the optimization algorithm converges to a stable solution. A multistart optimization strategy was adopted to handle the non-convex loss surface of the model, which can easily trap the optimization in suboptimal local minima.

To determine the optimal number of starts for the baseline fit, initial tests were run using 70 random initializations for each drug. Figure 9 shows the final mean squared error (MSE, y-axis) across the multistart runs (x-axis), with Pladienolide B in the upper panel and H3B-8800 in the lower panel. For both drugs, the lowest loss value plateaus after approximately 20 starts, showing that the optimizer reliably reaches a consistent minimum. To ensure convergence while keeping computational costs manageable, an empirical threshold of 35 starts was chosen for the global baseline fit.

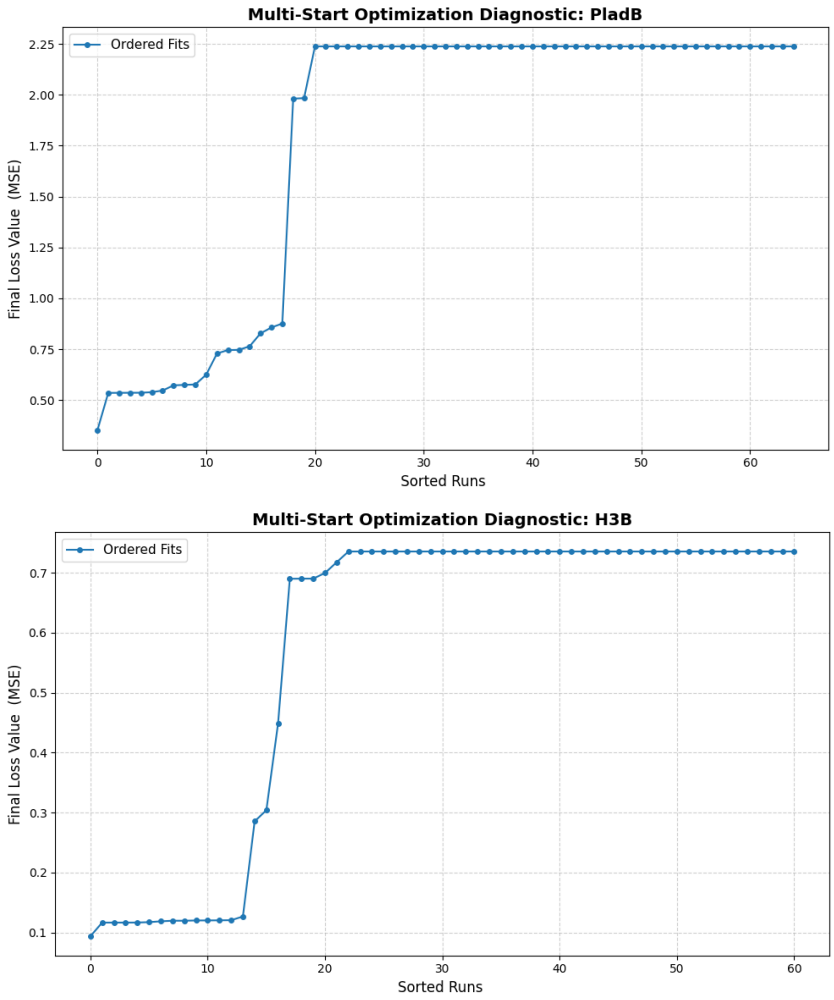


Figure 9 | **Multistart optimization convergence across random initializations.** Final mean squared error (MSE) loss values plotted against the number of multistart optimization runs for Pladienolide B (top panel) and H3B-8800 (bottom panel). The curve stabilizes at a plateau after ~20 starts for both

compounds, validating the choice of 35 multistart iterations as a safe and computationally efficient threshold for the global baseline fit.

3.3.2 Global and Intra-Class Fit Evaluation

Global evaluation of the model calibration results in high coefficients of determination (R^2) for both drugs: 0.84 for Pladienolide B and 0.85 for H3B-8800. This indicates that the kinetic model successfully explains most of the variance across the entire dataset.

To evaluate fit quality in detail, Table 3 compares the Root Mean Squared Error (RMSE) against the Experimental Range Length (the difference between maximum and minimum $\log_2$ FC values) across the four extreme architectural conditions.

Goodness of Fit — Pladienolide B

Condition	Experimental Range	Range Length	RMSE
Strong BP | Strong Decoy	[-2.90, 2.05]	4.95	0.29
Strong BP | No Decoy	[-3.55, 0.56]	4.11	0.59
Weak BP | Strong Decoy	[-1.56, 2.33]	3.89	0.10
Weak BP | No Decoy	[-1.24, 0.70]	1.93	0.01

Goodness of Fit — H3B-8800

Condition	Experimental Range	Range Length	RMSE
Strong BP | Strong Decoy	[-2.47, 2.37]	4.84	0.17
Strong BP | No Decoy	[-3.45, 0.37]	3.82	0.43
Weak BP | Strong Decoy	[-1.29, 2.64]	3.93	0.20
Weak BP | No Decoy	[-1.22, 1.14]	2.36	0.04

Table 3 | **Goodness of fit across extreme architectural regimes for Pladienolide B and H3B-8800.** Summary of model performance across the four extreme combinations of Branch Point (BP) and decoy site strength. Columns report the architectural condition, the experimental range [min, max] of log_2 FC, the total range length, and the resulting Root Mean Squared Error (RMSE) for each drug. The comparison highlights differences between dynamic, well-fitted regimes, regions of higher modeling residual, and flat, inert baselines.

The table shows 3 distinct patterns:

- **Strong Signal, Accurate Fit:** Conditions such as Strong BP | Strong Decoy and Weak BP | Strong Decoy shows a wide experimental range (e.g., 4.95 for PladB; 4.84 for H3B for the first condition) paired with a low RMSE (0.29 for PladB; 0.17 for H3B). This indicates that the drug effect drastically this conditions and the model is samehow capable of capturing this signal.
- **Strong Signal, Reduced Accuracy:** In the Strong BP | No Decoy condition, a wide empirical range (4.11 for PladB) is related to the highest observed error (RMSE of 0.59). This suggest that a real dose-dependent effect exists but the model difficulty reproduce the curve. This can be pointed out while examinating the condition specific parameters to see whether the parameters in this condition are particularly misinterpretable with the biology
- **Weak Signal, Correct Baseline:** Conditions like Weak BP | No Decoy display a narrow range length (1.93 for PladB) and a near-zero RMSE (0.01 for PladB). In this condition the experimental condition is biologically "flat" with minimal drug response. The model correctly predicts this lack of variation without artificially overfitting to local noise.

3.3.3 Validation of Boundary Kinetic Regimes

Given the quantitative evaluation of the goodness of fit, it is now possible to compare the macroscopic steady-state response with the empirical data across the different drug concentrations.

Figure 10 compares the empirical response against model predictions across drug concentrations. The top panel shows Pladienolide B and the bottom panel shows H3B-8800. From left to right, the subplots represent the four extreme regimes. In each plot, the x-axis indicates increasing drug concentrations and the y-axis reports the $\log_2$ Fold Change (LFC) relative to the control. Points and error bars represent the experimental mean and standard deviation (reflecting intra-condition sequence variability), while the solid line indicates the model prediction generated using the optimal estimated parameters.

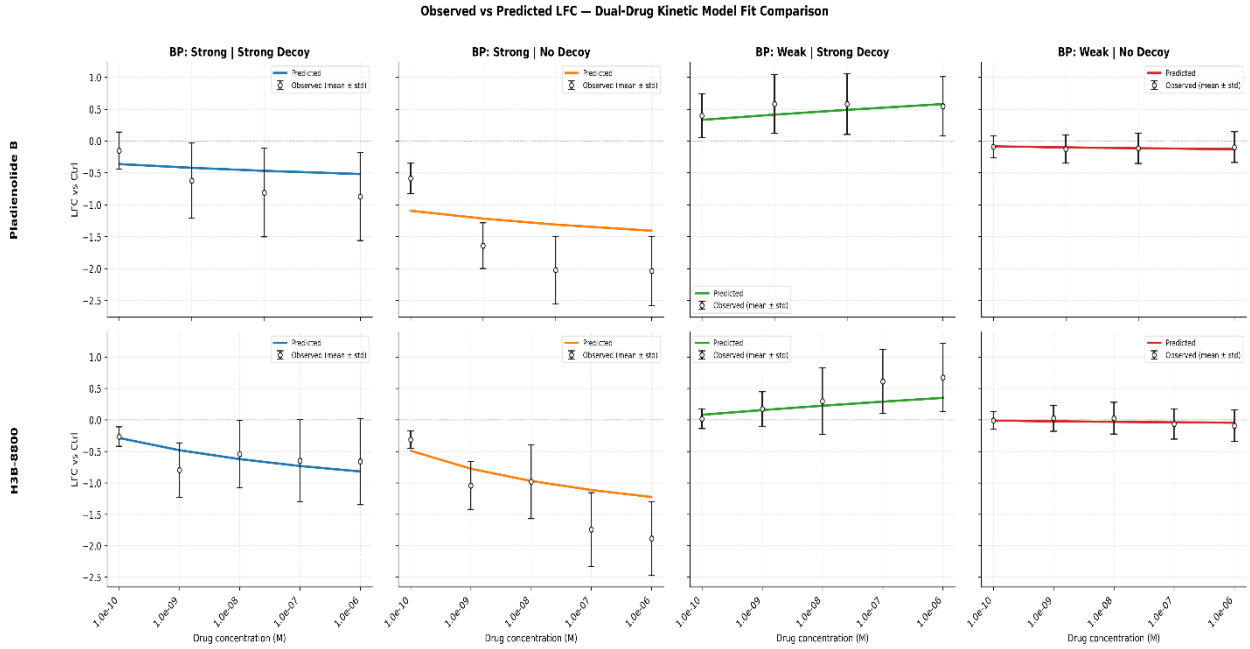


Figure 10 | **Empirical versus predicted dose-response trajectories across extreme sequence architectures.** Dose-response profiles showing log_2 Fold Change (LFC) relative to the untreated control across escalating drug concentrations for Pladienolide B (top panels) and H3B-8800 (bottom panels). Subplots from left to right display four extreme combinations: Strong BP / Strong Decoy, Strong BP / No Decoy, Weak BP / Strong Decoy, and Weak BP / No Decoy. Dots and vertical error bars denote empirical means and standard deviations across sequences within each class. Solid lines trace model predictions derived from optimal parameter fits.

An interesting observation is that the model's behavior is consistent across both independent drug profiles (PladB and H3B).

In conditions where the biological dynamics are simpler, such as with a weak Branch Point and a strong Decoy (e.g., BP_Weak / Decoy_Strong), the LFC increases almost linearly. Here, the model performs well and accurately overlaps the empirical means, which is supported by a very low RMSE of 0.29 for PladB and 0.17 for H3B across a wide experimental range length of 4.95 and 4.84, respectively.

In the baseline negative control condition (BP_Weak / No Decoy), where the biological response is flat, the model shows an almost near-zero RMSE of 0.01 (with a narrow range of 1.93 for PladB), correctly predicting this lack of variation without overfitting to local noise.

On the other hand, in conditions where the drug response is more non-linear (for instance, when the BP is strong), the model seems to perform worse. Even though the predicted line captures the overall decreasing slope starting from the first drug concentration, there is a clear discrepancy. This is particularly evident in the boundary classes designed without the decoy site (e.g.,

BP_Strong / No Decoy). In this case, across an empirical range of 4.11 for PladB, the model yields the highest observed error with an RMSE of 0.59: it still decreases following the empirical trend, but it struggles to fully resolve the steep curvature and variability of the data.

Overall, the model seems to perform accurately across both independent drugs. However, before concluding that the shared and condition-specific parameters are sufficient to capture the system, it is important to critically inspect these results. The central finding is that while the kinetic model captures the global structural trends under pharmacological stress, a simple two-parameter model has limitations in resolving these steep, condition-specific differences.

Mechanistic Interpretation and Rationale for Downstream Analysis

It is important to notice that the model seems to perform better on linear shapes rather than non-linear ones, despite being built on non-linear rational functions for the steady-state probability. The rationale behind this issue can be either supposed to be related to an incorrect model structure or 2 different technical limits of the experimental design:

- **Assay Resolution Limits:** Only a single experimental measurement of complex binding (Bound vs Unbound) is available. The assay cannot discriminate where the complex actually binds or its physical conformation (Open vs Closed). However, the model is trying to separate the strength of the two distinct sites (BP and Decoy) starting from this aggregated signal. This is a strong limitation for the identifiability of the kinetic rates.
- **Intra-Class Heterogeneity:** The internal variability of each experimental condition is high. This happens because the sequence strengths were computationally discretized (e.g., grouping them strictly into strong or weak BP classes). As a consequence, a single condition-specific parameter (β_c or δ_c) is forced to approximate an entire pool of sequences that still retain a lot of intrinsic biological variability.

To understand if these limitations result in actual structural non-identifiability, the analysis needs to move from a single point-estimate calibration to a proper quantification of the parameter's uncertainty as discussed in the Conclusions.

3.4 Parameter Uncertainty and Biological Identifiability

After the macroscopic validation of the model, it is necessary to evaluate the uncertainty of the estimated parameters and interpret those results in comparison with the biological expectations.

3.4.1 Bootstrap Implementation

In order to compute empirical Confidence Intervals (CIs) and quantify the variance of the estimates, a non-parametric bootstrap approach was implemented. For the final analysis the number of bootstrap iterations was set to 150 choosing this value as a trade off between statistical power and computational cost (the simulation time for this in silico set up was rougly 2 hours for each drug)

During initial trials of the simulation to reduce the computational time were implemented fewer number of optimization start ($n = 15$) and the final loss values across the bootstrap replicates showed a bimodal distribuction with one cluster falling around ~ 0.47 and another around ~ 1.13 for H3B. This indicates that the $n = 15$ will lead to a 2 different local minima and will introduce a computational artifact suggesting also the importance of a multistart algorithm for this specific problem. Coherentely with the plot shown in Fig. 9 (section 3.3.1) it was choosen a multistart iteration number of $n = 35$ leading to a stable cluster of loss of around ~ 0.36.

3.4.2 Cross-Drug Validation and Parameter Overview

Once the bootstrap distributions were obtained independently for both pharmacological treatments, the resulting CI and full parameter distributions were compared. The rationale behind this cross-drug comparison is to test whether the parameter ranges are coherent across both independent datasets (PladB and H3B), and then evaluate if this coherence is biologically meaningful.

The following sections examine the biological expectation and than how the statistical distributions of these parameters compare against those, separating the analysis into the shared kinetic parameters and the condition-specific scalers.

3.4.3 Kinetic Assumptions and Parameter Expectations

The kinetic parameters of the model are constrained by specific biological expectations regarding the spliceosome assembly pathway. The basal dissociation rate from the Decoy site (k_{-2}) is expected to be greater than the dissociation from the Branch Point in the open conformation (k_{-1}). The rationale behind this expectation is that, under control conditions and regardless of sequence strength, the SF3B1 complex is less likely to remain stably bound to a decoy site compared to a functional Branch Point. The transition rate representing the conformational change from the open to the closed state at the Branch Point (k_3) is hypothesized to be extremely fast. This is because it represents an irreversible ATP-dependent step that is faster than reversible binding. The dissociation rate from the fully closed conformation (k_4) instead should be very low, reflecting the stability of the clamped state.

Regarding the condition-specific scaling parameters (β and δ), their design constrains them within the [0,1] interval, acting as linear multipliers of the basal dissociation constants. Then, the strongest cis-regulatory elements (for both the BP and the Decoy) are expected to be closer to 0. The rationale is that a near-zero multiplier minimizes the dissociation rate, meaning that the complex is more likely to remain bound to strong sequences at the steady state.

3.4.4 Shared Kinetic Parameters

Figure 11 shows the bootstrap distributions for the four free structural rates ($k_{-1}^{(0)}, k_{-2}^{(0)}, k_3^{(0)}, k_4$) describing complex binding and conformational transitions. The association rates k_1 and k_2 are not shown because they were set to 1 to set a baseline scale for the model. The drug sensitivity parameter α was estimated during the joint fit, but it is not analyzed here because its role is only to scale the drug concentration for each compound, whereas the focus of this section is on the structural properties of SF3B1 itself. In the plot, parameters are displayed along the x-axis, comparing Pladienolide B and H3B-8800 distributions side by side, with estimated rate values reported on the y-axis. Whiskers and box boundaries indicate confidence intervals and interquartile ranges across resampled datasets, while brackets report pairwise statistical comparisons evaluated via the Mann-Whitney U test (annotated as *ns* for non-significant differences).

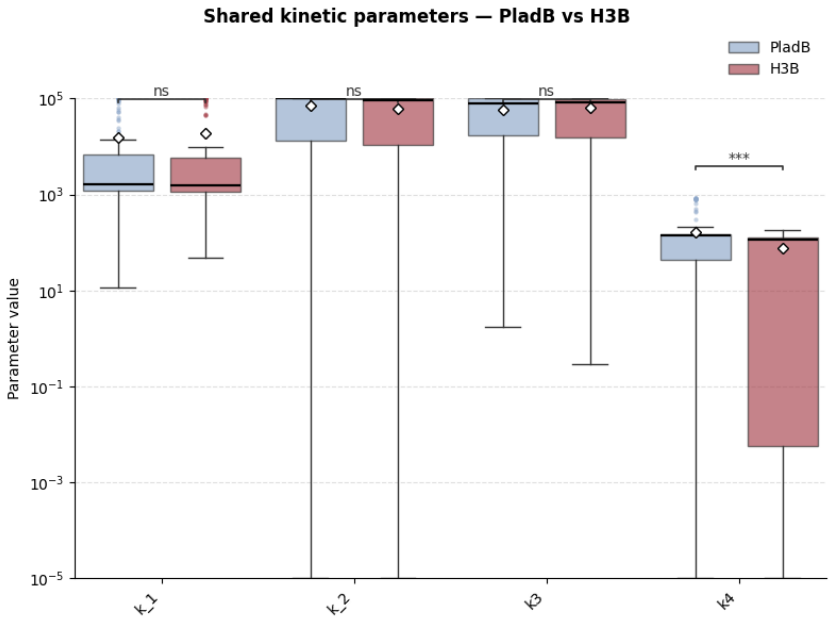


Figure 11 | **Bootstrap uncertainty distributions and cross-drug comparison of shared kinetic parameters.** Boxplots showing the parameter estimates obtained across bootstrap resamplings for the shared kinetic rates governing SF3B1 binding and conformational transitions. Values for Pladienolide B and H3B-8800 are compared side by side. Horizontal brackets denote statistical significance assessed by two-sided Mann-Whitney U tests (ns: $p \geq 0.05$). Dashed horizontal lines indicate the lower and upper bounds of the optimization parameter space, highlighting parameter identifiability and boundary-stacking behaviors.

Across statistical, mathematical, and biological perspectives, a few trends arise:

- **Statistical Perspective (CI Width and Overlap):**

For most of the shared parameters (k_{-1}, k_{-2}, k_3), the bootstrap distributions show narrow confidence intervals, at least from the 1st quartile to the maximum, spanning about two orders of magnitude. However, k_{-2} has spread tails below the 1st quartile. This spread indicates a discrete estimation variance across the resampled datasets, with more uncertainty associated with the latter parameter. Within the comparison of PladB and H3B using the Mann-Whitney U test, the distributions show no statistically significant differences, marked as *ns* (not significant). This overlap provides consistency to the estimates. On the other hand, the k_4 parameter shows significant differences between the two drugs, but with a comparable median and a very broad CI for the H3B condition which may suggest still that the 2 estimation are consistent.

- **Mathematical Perspective (Boundary Stacking):**

A mathematical issue is that k_{-2} and k_3 are stretched to the upper limits of the optimization boundaries, with k_{-2} particularly covering the whole

optimization space. These stretched parameters and wide distributions suggest a lack of information in the data to precisely constrain that rate, which refers to a non-identifiability problem.

- **Biological Interpretation vs. Empirical Identifiability**

Basal Dissociation Rates (k_{-1} vs k_{-2}): Biologically, the Decoy dissociation rate (k_{-2}) was expected to be consistently higher than the Branch Point open dissociation rate (k_{-1}). This expectation is consistent when considering the data above the 1st quartile, but there is still a wide range of data points that are inconsistent with this hierarchy.

Conformational Switch and Stable Dissociation (k_3 and k_4): The expectation for k_3 (a rapid ATP-dependent transition) can be recovered in its tendency to be stretched to the upper mathematical boundaries, but because of this boundary stacking, it cannot be strictly interpreted as being definitively greater than the other rates. However, the dissociation rate from the closed state (k_4) remains at much lower values, aligning with the biological expectation of a stable complex.

The fact that the parameters overlap consistently between the two independent drugs suggests that, at this stage of the analysis, the model seems to correctly represent the underlying biology.

Overall, these parameters indicate that the kinetic model is able to capture the global trends and confirms the presence of a stable biological scaffold (represented by these parameters that seems to be biologically meaningful).

3.4.5 Condition Specific Parameters

While the shared parameters describe the global kinetic, the condition-specific scalers (δ and β) are designed to capture the differencies inside the sequences. As defined in the model construction, these parameters are constrained within the [0,1] interval and act as linear multipliers for the basal dissociation rates k_{-2} (Decoy) and k_{-1} (Branch Point).

Looking at the bootstrap distributions across the biological conditions, it is clear that the estimated parameters diverge from this expectation.

The Decoy Scaler (δ)

Figure 12 illustrates the bootstrap distributions of the δ parameter across the four architectural conditions. In the plot, conditions are arranged along the x-axis, with Pladienolide B and H3B-8800 shown side by side for direct

comparison. The y-axis represents the scaler value bounded in the [0,1] interval. Brackets and asterisks indicate pairwise statistical differences evaluated via the Mann-Whitney U test.

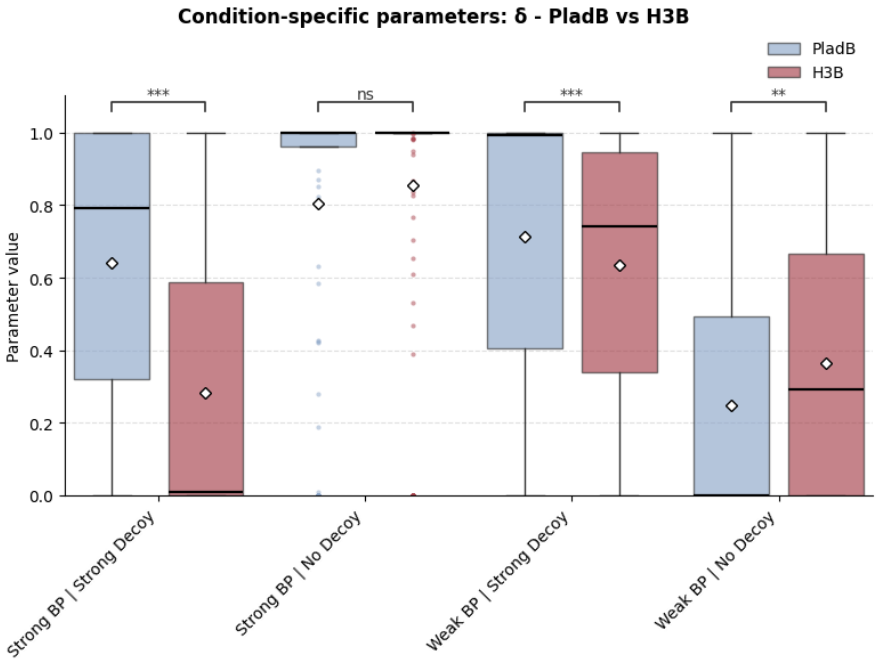


Figure 12 | **Bootstrap distributions of the decoy site dissociation scaler (δ) across architectural regimes.** Boxplots showing parameter estimates for β (bounded in [0,1]) across bootstrap iterations for Pladienolide B and H3B-8800. Conditions on the horizontal axis correspond to combinations of Branch Point and Decoy site strengths. Brackets denote statistical significance between drugs (Mann-Whitney U test; ***: $p < 0.001$). Horizontal dashed lines mark the parameter boundaries (0 and 1), illustrating the widespread boundary-stacking behavior and divergence from biological expectations in strong decoy contexts.

- **Mathematical and Statistical Perspective:** The confidence intervals are extremely wide, often covering the entire [0,1] sample space. Significant differencies (***) are also observed between Pladienolide B and H3B-8800 in almost all classes. The parameters are also streched at the mathematical boundaries.
- **Biological Perspective:** In the "No Decoy" conditions, the biological expectation is that δ should be close to 1 because without a decoy site any binding should dissociate with the basal rate. The model partially respects this in the Strong BP | No Decoy class, where δ stacks at 1. However, in the Strong Decoy conditions, the expectation is that δ should be close to 0. Instead, contradictory behaviors are found: for PladB in Strong BP | Strong Decoy, the median is around 0.8, and in Weak BP | Strong Decoy it stacks exactly at 1. This means the model is

mathematically treating strong decoy sites as if they were weak or non-existent.

The Branch Point Scaler (β)

Figure 13 displays the bootstrap distributions for the β parameter across the same architectural combinations. As in Figure 12, the conditions are organized along the horizontal axis, contrasting the two drug treatments side by side across the bounded range [0,1] on the vertical axis.

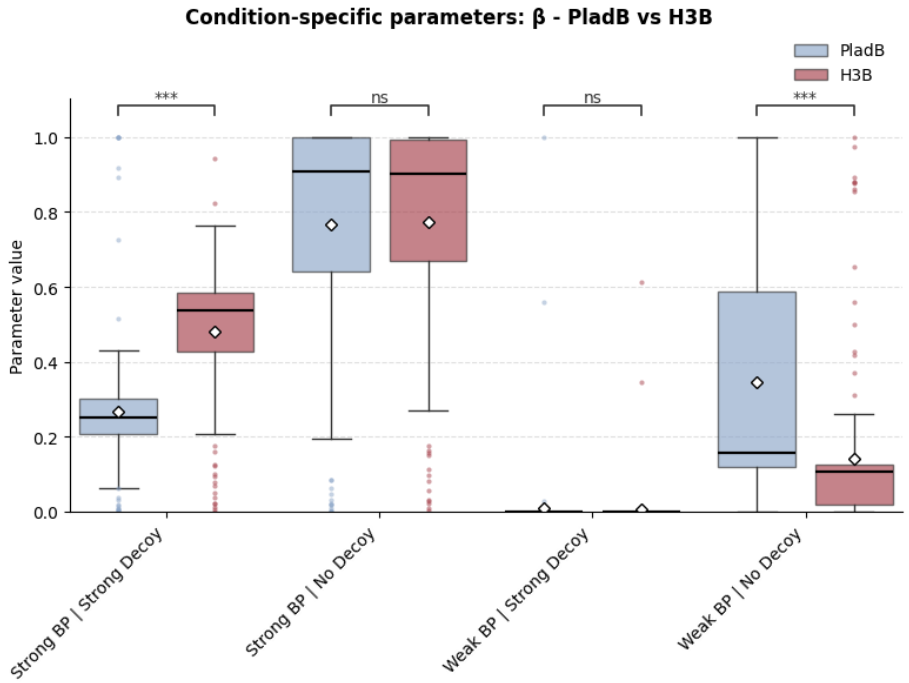


Figure 13 **| Bootstrap distributions of the branch point dissociation scaler (β) across architectural regimes.** Boxplots of bootstrap parameter estimates for the open branch point dissociation scaler (β) within the [0,1] bounded interval, comparing Pladienolide B and H3B-8800 across sequence architectures. The plots highlight structural non-identifiability and inverted biological trends, evidenced by boundary stacking at 0 for weak BP regimes and at 1 for strong BP constructs lacking decoy sites.

- **Mathematical and Statistical Perspective:** Similar to δ, are present boundary stacking. For instance, the Weak BP | Strong Decoy class collapses to 0 for both drugs. In contrast, the Strong BP | No Decoy class stacks to the upper boundary of 1.
- **Biological Perspective:** Also here the mathematical fit diverge from the biological reality. A Strong BP should have a β near 0. While the Strong

BP | Strong Decoy class shows moderately low medians (~0.25 for PladB and ~0.5 for H3B), the Strong BP | No Decoy class peaks near 1, which is the exact opposite of what a strong Branch Point should do. Also the Weak BP classes, which should have β values near 1, tend to be closer to 0, incoherently with the biological expectation.

Taken together, these results indicate that while the shared parameters display a biologically coherence across both drugs, the condition-specific scalers β and δ fail to reflect the expectation of site strength.

Whether this reflects a limitation of the data, an identifiability issue, or a structural gap in the model itself is discussed in the Conclusion, together with possible strategies to progress further on these scenarios.

4 Conclusions and Future Perspectives

In this work, a mechanistic kinetic model was developed to describe the competitive recruitment of the SF3B1 spliceosomal complex between two alternative cis-regulatory elements (the functional Branch Point and an upstream Decoy site) under pharmacological perturbation by two spliceostatin-family inhibitors (Pladienolide B and H3B-8800). By using the linear framework for biochemical systems, the steady-state occupancy probabilities across four discrete microstates ($S_0, S_D, S_{\text{open}}, S_{\text{closed}}$) were derived using the Matrix-Tree Theorem. This allowed to calculate directly the steady state of the network without having to solve ODE equations numerically during the fitting.

The Global Shared Parameters ($\Phi = \left\{k_{-1}^{(0)}, k_{-2}^{(0)}, k_3^{(0)}, k_4, \alpha\right\}$, with association rates set to $k_1 = k_2 = 1$) represent the sequence-independent kinetic scaffold of basal binding, conformational transitions, and pharmacological inhibition, and are kept invariant across the entire library.

The Condition-Specific Scalers (β and δ), constrained in the unit interval [0,1], act instead as linear multipliers on the basal dissociation rates ($k_{-1} = \beta \cdot k_{-1}^{(0)}$ and $k_{-2} = \delta \cdot k_{-2}^{(0)}$) to encode the intrinsic strength of the Branch Point and Decoy site, respectively. To quantify parameter uncertainty without relying on parametric assumptions, non-parametric bootstrap distributions (B=150) and empirical 95% confidence intervals were evaluated independently for both pharmacological series.

The evaluation of the fitted parameters reveals two distinct regimes. For the shared kinetic rates ($k_{-1}^{(0)}, k_{-2}^{(0)}, k_3^{(0)}$), the bootstrap distributions display narrow confidence intervals, and a two-sided Mann-Whitney U test confirms that these estimates show no statistically significant differences ($p \geq 0.05$) between the PladB and H3B datasets. The k_4 parameter is the only exception, showing a statistically significant difference between the two drug treatments, although its median values remain very close across the two series. Overall, the estimated shared parameters are consistent with the biological expectations: the basal dissociation rate from the branch point is lower than from the decoy site ($k_{-1}^{(0)} < k_{-2}^{(0)}$), which makes biological sense, since in the control condition SF3B1 forms a more stable interaction with a functional branch point than with a decoy site. The forward transition rate to the closed state ($k_3^{(0)}$), is very fast and reaches high values, consistent with an irreversible, ATP-dependent conformational change that locks the complex in place; its magnitude is nonetheless close to that of $k_{-2}^{(0)}$, which is somewhat less expected, since the model assumes this forward transition to be considerably faster than the basal

off-rates. The dissociation rate from the closed state (k_4) is very low compared to all other rates ($k_4 \ll k_{-1}^{(0)}$), matching the biology well: because the closed conformation is extremely stable, unbinding from this locked state is strongly disfavored.

The condition-specific scalers, on the other hand, reveal clear limitations. Across the four extreme boundary regimes, the two drug datasets show low consistency: the Decoy scaler (δ) displays comparable confidence intervals in 3 out of 4 boundary conditions, while the Branch Point scaler (β) remains statistically similar in only 2 out of 4. More critically, the numerical values do not follow the expected biological hierarchy ($\beta_{Strong} < \beta_{Intermediate} < \beta_{Weak}$ and $\delta_{Strong} < \delta_{Weak}$): strong regulatory elements, which should converge toward lower scaler values, more frequently approach the upper bound of the parameter space instead, and vice versa for the weak strengths. From a statistical standpoint, the resulting confidence intervals are wide, at times occupying the entire [0,1] parameter space, suggesting that the current data and model formulation lack sufficient information to constrain the sequence-specific scalers.

Several alternative explanations can explain why the condition-specific scalers β and δ fail to follow the expected biological hierarchy, ranging from limitations in the data to possible simplifications in the model structure itself.

A first possibility is practical non-identifiability, caused by the limitations of the data. The experimental readout is aggregated: it is only known whether the RNA is bound or unbound to the SF3B1 complex, not whether the complex is bound at the Branch Point or at the Decoy, nor whether it sits in an open or closed state. Combined with the limited number of drug concentration points available (four for PladB and five for H3B), the experimental data may simply be uninformative enough to discriminate the individual effects of the two parameters.

A second, more fundamental possibility is structural non-identifiability, intrinsic to the mathematical formulation itself. In the steady-state formulas, β and δ might be coupled such that different combinations of these parameters produce the same predicted bound fraction; if so, the optimizer has no way of converging to a unique solution. Verifying this would require a formal structural identifiability analysis to establish whether the two parameters can, in principle, be separated whithin this mathematical structure.

Even if the fitting were both numerically and structurally correct, a third possibility is that the biological reading of the parameters is too simplistic. It was assumed an intuitive rule where a strong sequence must correspond to a scaler close to zero, but the parameters of a kinetic network interact with one

another rather than acting in isolation: the very fast transition to the closed state ($k_3^{(0)}$), for instance, might dominate the system's dynamics and compensate for changes in the dissociation rates. Under this scenario, the fitted β and δ values could be mathematically coherent, yet not interpretable through a simple "closer to zero means stronger" rule.

Finally, at a structural level, the four-state topology itself might be incomplete. Reducing the complex differences in sequence affinity to two simple linear multipliers may be an oversimplification: the real biological mechanism could involve additional steps not considered here, such as cooperative binding, kinetic proofreading, or a distinct conformational switch. If the true system is more complex than assumed, the four-state graph or the only two parameter setup, may simply be incapable of capturing the sequence-specific differences with accuracy.

Previous models (Donovan et al., 2024) have described the early steps of spliceosome assembly by focusing on how factors bind to a single, isolated pre-mRNA sequence. The main contribution of this thesis is the shift from an isolated-binding perspective to one of site-specific competition: by considering simultaneously the canonical Branch Point and the upstream Decoy site competing for the same available pool of SF3B1 complex, the model captures a more realistic spatial dynamic of assembly. This competition is also coupled with the open-to-closed conformational switch, allowing the model to represent the step targeted by spliceostatins. Overall, this approach suggests that incorporating active competition between alternative binding sites can offer a more realistic view of spliceosome behavior under pharmacological stress, compared to models based solely on simple binding affinities.

Improving this model requires addressing the remaining experimental and methodological limitations. As already discussed, the aggregated nature of the readout and the limited number of drug concentration points make it difficult to precisely estimate the condition-specific parameters. Beyond this, two further limitations should be noted. First, the continuous Branch Point strengths were grouped into broad discrete categories: using a single parameter for an entire group inevitably discards the real biological differences between individual sequences within that class. Second, the dataset relies on a single experimental replicate (n=1); although the massive size of the library provides substantial statistical power at the architectural level, true biological replicates remain necessary to properly quantify experimental noise.

Addressing these issues will require improvements on both the computational and experimental fronts. Computationally, Physics-Informed Neural Networks (PINNs) could estimate the kinetic parameters (β_s, δ_s) directly for each

individual RNA sequence, removing the need to group them into broad strength classes. In this setup, a neural network predicts the kinetic rates from the sequence itself, while the equations of the kinetic model guide the learning process. This would allow the model to run on single sequences using their specific parameters, combining machine learning flexibility with the biological rules of the system.

Experimentally, higher-resolution assays could resolve the current identifiability limitations: techniques such as CLIP-seq or footprinting could reveal exactly where the complex binds, while a heparin challenge could help physically separate open from closed complexes. Finally, a Profile Likelihood analysis would help clarify whether the parameter issues come from having too few data points or from the model equations themselves. While the bootstrap only measures how much estimates vary across resampled data, Profile Likelihood tests one parameter at a time across different values while re-fitting the rest. This shows whether the data contains a clear minimum (practical identifiability) or whether the loss stays completely flat because two parameters cancel each other out (structural identifiability).

As noted in Materials and Methods, this work focused on parameter estimation and identifiability rather than predictive performance. A natural next step is therefore to split the data into training and test sets, so that the model's predictive accuracy and its tendency to overfit can be estimated.

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